

Western MA Per Pupil Expenditure Report

With selected districts for comparison

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Appendix A. Data Sources & Calculation Methodology

1. Data sources

Expenditure dataset

Massachusetts Department of Elementary and Secondary Education (DESE)

District Expenditures by Spending Category

Last updated: August 12, 2025

URL: <https://educationtocareer.data.mass.gov/Finance-and-Budget/District-Expenditures-by-Spending-Category/er3w-dyti/>

This dataset provides:

- Expenditures Per Pupil by category for all Massachusetts school districts
- Student enrollment (In-District FTE, Out-of-District FTE, Total FTE)
- Annual End-of-year-report (EOYR) data from FY2009 to present

Chapter 70 District Profiles

Massachusetts Department of Elementary and Secondary Education (DESE)

Chapter 70 District Profiles

Retrieved: September 2025

URL: <https://www.doe.mass.edu/finance/chapter70/default.html>

From the Ch70 website: "Chapter 70 District Profiles: The on-line Chapter 70 database shows, for each school district, yearly spending and state aid totals in comparison to the foundation budget. Trend data is available for each year going back to FY1993."

This dataset provides:

- Chapter 70 Aid (c70aid): State aid received by each district
- Required Net School Spending (rqdnss2): Minimum spending required by state law
- Actual Net School Spending (actualNSS): Total district spending on education
- Foundation Enrollment (distfoundenro): District foundation enrollment used for Ch70 funding calculations
- Data reported annually each Fall from FY1993 to present

District classifications and geography

MassGIS Data: Public School Districts

Polygon data layers depicting the boundaries of public school districts in the Commonwealth of Massachusetts.

School type classifications (Traditional, Regional, Charter, etc.)

Last updated: June 2019

URL: <https://www.mass.gov/info-details/massgis-data-public-school-districts>

MassGIS Data: MA Executive Office of Health & Human Services Regions

Districts classified by region based on EOHHS regional designations.

Last updated: April 2022

URL: <https://www.mass.gov/info-details/massgis-data-ma-executive-office-of-health-human-services-regions>

2. Key definitions & calculations

Per-pupil expenditure (PPE)

PPE is reported by the DESE EOYR. It is calculated with In-District FTE enrollment in the denominator. Per DESE's Researcher's Guide: "Out-of-district costs cannot be properly reported as per-pupil expenditure because tuition costs vary greatly depending on the reason for going out of district."

PPE includes 10 subcategories in the EOYR:

- Administration
- Guidance, Counseling and Testing
- Instructional Leadership

- Instructional Materials, Equipment and Technology
- Insurance, Retirement Programs and Other
- Operations and Maintenance
- Other Teaching Services
- Professional Development
- Pupil Services
- Teachers

This report groups two subcategories in 'Other':

- Instructional Materials, Equipment and Technology
- Professional Development

Chapter 70 and Net school spending (NSS)

To understand funding sources, we decompose total spending into three stacked components:

- Chapter 70 Aid (\$/pupil): $\text{State funding} = \text{c70aid} / \text{foundation enrollment}$
- Required local contribution (\$/pupil) = $\max(0, \text{rqdnss2} - \text{c70aid}) / \text{foundation enrollment}$
- Actual local contribution above required (\$/pupil) = $(\text{actualNSS} - \text{rqdnss2}) / \text{foundation enrollment}$

Total NSS per foundation pupil = Sum of all three components = $\text{actualNSS} / \text{foundation enrollment}$

Note about enrollment denominators

Chapter 70 aid and net school spending calculations use foundation enrollment for per-pupil figures, while PPE totals and expense categories use in-district FTE enrollment. These metrics come from different DESE reporting touchpoints:

- Foundation enrollment (Chapter 70 profile): All students residing in the district on October 1st of the prior year for whom the district is financially responsible.
- In-district FTE (End-of-year-report (EOYR)): Full-year enrollment calculated from total membership days for students either residing in the district, or choiced or tuitioned into the district.

This difference complicates direct comparison between Chapter 70 and expense analyses, but remains valid since each district uses the same denominator type within each metric. The comparison is not apples-to-oranges, but a picture of apples in the morning compared with a picture of apples in the evening. For more information, see the [DESE Researcher's Guide](https://www.doe.mass.edu/research/researchers.html): <https://www.doe.mass.edu/research/researchers.html>

Compound Annual Growth Rate (CAGR)

CAGR measures the smoothed annual growth rate over multiple years — answering "what steady annual rate would produce the same start-to-end result?"

Formula: $\text{CAGR} = (\text{End_Value} / \text{Start_Value})^{(1 / \text{Years})} - 1$

Example: If expenditures grow from \$10,000 to \$15,000 over 10 years:

$\text{CAGR} = (\$15,000 / \$10,000)^{(1/10)} - 1 = 4.14\% \text{ per year}$

Weighted average

For cohort aggregates, we use enrollment-weighted averages to ensure larger districts appropriately influence the cohort average:

Formula: $\text{Weighted_PPE} = \Sigma(\text{District_PPE} \times \text{District_FTE}) / \Sigma(\text{District_FTE})$

3. Enrollment-based cohort methodology

Districts are grouped by FY2024 In-District FTE enrollment using quartile analysis:

Cohort boundaries (FY2024)

Based on 59 Western MA traditional districts:

- Q1 (25th percentile): 154.7 FTE → rounded to 200
- Median (50th percentile): 792.0 FTE → rounded to 800
- Q3 (75th percentile): 1,597.6 FTE → rounded to 1,600
- Outlier threshold: Fixed at 10,000 FTE

Final cohorts:

- **Tiny (0-200 FTE):** 17 districts (Cohort 1: below Q1) **Member districts:** conway, erving, farmington river reg, florida, hancock, hawlemont, leverett, new salem-wendell, pelham, petersham, richmond, rowe, savoy, shutesbury, sunderland, and 2 others
- **Small (201-800 FTE):** 13 districts (Cohort 2: Q1 to median) **Member districts:** clarksburg, deerfield, frontier, gateway, granby, hadley, hatfield, lee, lenox, orange, pioneer valley, ralph c mahar, southern berkshire
- **Medium (801-1600 FTE):** 15 districts (Cohort 3: median to Q3) **Member districts:** amherst, amherst-pelham, berkshire hills, central berkshire, easthampton, gill-montague, greenfield, hoosac valley regional, mohawk trail, monson, mount greylock, north adams, palmer, southwick-tolland-granville regional school district, ware
- **Large (1601-10K FTE):** 14 districts (Cohort 4: above Q3) **Member districts:** agawam, athol-royalston, belchertown, chicopee, east longmeadow, hampden-wilbraham, holyoke, longmeadow, ludlow, northampton, pittsfield, south hadley, west springfield, westfield
- **Outliers (Springfield >10K FTE):** 1 district (Cohort 5: statistical outlier, analyzed separately) **Member districts:** springfield

4. Comparison shading methodology

To visualize meaningful differences while filtering statistical noise, we use graduated color shading based on two independent tests:

Colors

- Orange: District exceeds baseline
- Blue: District below baseline
- White: Within threshold (not meaningfully different)

Test 1: Dollar amounts (2024 values)

- Threshold: $\geq 5.0\%$ difference from baseline
- Calculation: $(\text{District} - \text{Baseline}) / \text{Baseline}$
- Shading intensity bins: 5%, 10%, 15%, 20%+

Test 2: Growth rates (CAGR)

- Threshold: ≥ 0.5 percentage points (pp) difference from baseline
- Calculation: $\text{District_CAGR} - \text{Baseline_CAGR}$
- Shading intensity bins: 0pp, 1pp, 2pp, 2pp+

Statistical justification for thresholds

Analysis of all 59 Western MA districts revealed different natural variation by metric:

- PPE variation: Mean = \$24,237, SD = \$5,462, CV = 22.5%
- CAGR variation: Mean = 6.00pp, SD = 3.24pp, CV = 54.0%

The 5.0%/0.5pp thresholds balance statistical rigor with practical communication:

- 5.0% for PPE: Flags differences $\geq \$1,212$ at mean (~ 0.22 SD)
- 0.5pp for CAGR: Flags differences $\geq 0.5\text{pp}$ (~ 0.15 SD)
- Balance ratio: $0.7\times$ (appropriately sensitive to compound growth effects)

This approach creates a visual hierarchy:

- No shading: Districts statistically similar to peers
- Lighter shading: Notable difference
- Darker shading: Exceptional difference requiring attention

These thresholds reflect the different ways these metrics behave. PPE and enrollment are dollar amounts and headcounts — a

5.0% difference is substantial and visible in annual budgets. But CAGR measures rates of change, and small differences in growth rates compound dramatically over time. Consider a district with \$20,000 PPE:

- Growing at 4.0% annually reaches \$36,019 after 15 years
- Growing at 4.5% annually reaches \$38,318 after 15 years
- That 0.5pp difference produces a \$2,299 gap (11.5% more total growth)

For a 300-student district, that's \$689,700 in additional annual spending by year 15. For municipalities managing long-term budgets and multi-year labor contracts, these trajectories matter—even when year-to-year changes seem modest.

The tighter CAGR threshold (0.5pp vs. 5%) also reflects statistical reality: growth rates vary more widely across districts (CV = 54%) than PPE levels do (CV = 22.5%). A 0.5pp threshold flags districts with growth trajectories meaningfully different from their cohort without overcounting normal variation. By using a more sensitive threshold for CAGR, the analysis captures the real-world policy significance of compound growth differences while maintaining appropriate sensitivity for dollar and enrollment comparisons.

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Appendix B. Calculations and Examples

Complete calculations and worked examples for all figures and tables

Purpose

This appendix provides calculations and worked examples for all figures and tables in the report. It combines step-by-step calculation procedures with detailed examples for verification.

Figure 1: Year-over-Year (YoY) Growth Rates

Location: Executive Summary page

Source: executive_summary_plots.py, lines 44-63

Formula:

$$\text{YoY Growth}(\text{year}) = [(PPE(\text{year}) / PPE(\text{year}-1)) - 1] \times 100$$

Calculation Steps:

1. For each district, get total PPE by year:

- Load expenditure data from df (all expense categories)
- Pivot to get PPE by category and year
- Sum across all categories: $\text{total_ppe}[\text{year}] = \text{sum}(PPE_category[\text{year}] \text{ for all categories})$

2. Calculate YoY growth:

- For year in [2010, 2011, ..., 2024]:

$$\text{YoY}[\text{year}] = [(\text{total_ppe}[\text{year}] / \text{total_ppe}[\text{year}-1]) - 1] \times 100$$

- First year (2009) has no YoY value (no prior year)

3. For cohort aggregates:

- Get all districts in cohort
- Calculate weighted aggregate PPE (see Calculation #5)
- Apply same YoY formula to aggregate PPE series

Example (Amherst 2023-2024):

- $PPE(2023) = \$25,432.18$
- $PPE(2024) = \$26,789.45$
- $\text{YoY Growth} = [(26,789.45 / 25,432.18) - 1] \times 100 = 5.34\%$

Figure 2: 5-Year CAGR (Compound Annual Growth Rate) - Grouped Bars

Location: Executive Summary page

Source: executive_summary_plots.py, lines 66-95

Formula:

$$\text{CAGR} = [(\text{End_Value} / \text{Start_Value})^{(1/\text{Years})} - 1] \times 100$$

Calculation Steps:

1. Define three 5-year periods:

- Period 1: 2009-2014 (5 years)

- Period 2: 2014-2019 (5 years)
 - Period 3: 2019-2024 (5 years)
2. For each district and each period:
- Get total_ppe[start_year] and total_ppe[end_year]
 - $CAGR = [(total_ppe[end] / total_ppe[start])^{(1/5)} - 1] \times 100$
3. For cohort aggregates:
- Use weighted aggregate PPE (see Calculation #5)
 - Apply same CAGR formula

Example (Medium Cohort 2019-2024):

- Weighted PPE(2019) = \$22,145.67
- Weighted PPE(2024) = \$28,934.21
- $CAGR = [(28,934.21 / 22,145.67)^{(1/5)} - 1] \times 100$
- $CAGR = [1.30672^{0.2} - 1] \times 100$
- $CAGR = [1.05497 - 1] \times 100 = 5.50\%$

Figure 3: 15-Year CAGR (2009-2024)

Location: Executive Summary page

Source: executive_summary_plots.py, lines 98-118

Formula:

$$CAGR_{15yr} = [(PPE(2024) / PPE(2009))^{(1/15)} - 1] \times 100$$

Calculation Steps:

1. Get total_ppe[2009] and total_ppe[2024] for each district
2. Apply CAGR formula with years = 15
3. For cohort aggregates, use weighted aggregate PPE

Example (Amherst):

- PPE(2009) = \$18,234.56
- PPE(2024) = \$26,789.45
- $CAGR = [(26,789.45 / 18,234.56)^{(1/15)} - 1] \times 100$
- $CAGR = [1.46923^{0.06667} - 1] \times 100 = 2.61\%$

2. COHORT DETERMINATION CALCULATIONS

Purpose: Assign each district to one of 6 enrollment-based cohorts

Source: school_shared.py, lines 140-200 (calculate_cohort_boundaries)

Cohort Definitions:

- TINY: 0 - Q1 (25th percentile)
- SMALL: Q1 - Q2 (median)
- MEDIUM: Q2 - Q3 (75th percentile)
- LARGE: Q3 - 10,000 FTE

- SPRINGFIELD: > 10,000 FTE (outlier district)

Calculation Steps (FY2024):

1. Get all Western MA traditional districts

2. Get FY2024 FTE enrollment for each district

3. Create array of all enrollments (INCLUDING Springfield):

all_enrollments = [154.7, 287.3, ..., 24,567.8] # 59 districts

4. Calculate percentiles ON FULL DATASET:

- Q1 (25th) = np.percentile(all_enrollments, 25) = 154.7

- Q2 (50th) = np.percentile(all_enrollments, 50) = 520.3

- Q3 (75th) = np.percentile(all_enrollments, 75) = 1,597.6

- P90 (90th) = np.percentile(all_enrollments, 90) = 3,892.1

5. Round to clean boundaries:

- Q1: 154.7 → 200 FTE

- Median: 520.3 → 500 FTE

- Q3: 1,597.6 → 1,600 FTE

- P90: 3,892.1 → 4,000 FTE

- Outlier threshold: Fixed at 10,000 FTE

6. Final FY2024 Cohort Boundaries:

- TINY: 0-200 FTE (13 districts)

- SMALL: 201-500 FTE (16 districts)

- MEDIUM: 501-1,600 FTE (15 districts)

- LARGE: 1,601-10,000 FTE (14 districts)

- SPRINGFIELD: >10,000 FTE (1 district: 24,567.8 FTE)

Note: For historical data, cohorts are calculated year-by-year using that year's enrollment distribution.

3. WEIGHTED AGGREGATION METHODOLOGY

Purpose: Calculate enrollment-weighted per-pupil expenditure for a group of districts

Source: school_shared.py, lines 1001-1050 (weighted_epp_aggregation)

Formula:

Weighted PPE(category, year) = $\sum [PPE(d, cat, yr) \times FTE(d, yr)] / \sum [FTE(d, yr)]$

where d = district, cat = category, yr = year

Calculation Steps:

1. For each district d in cohort:

- Get PPE by category and year: PPE(d, category, year)

- Get FTE enrollment by year: FTE(d, year)

2. For each category and year:

- Calculate weighted numerator:

numerator(cat, yr) = $\sum [PPE(d, cat, yr) \times FTE(d, yr)]$ for all d in cohort

- Calculate total FTE:

$\text{denominator}(\text{yr}) = \sum [\text{FTE}(\text{d}, \text{yr})]$ for all d in cohort

- Weighted PPE(cat, yr) = numerator(cat, yr) / denominator(yr)

Example (Medium Cohort, Instruction category, 2024):

Districts in Medium cohort (FY2024): 15 districts

District A: PPE_instruction = \$12,345, FTE = 678

District B: PPE_instruction = \$13,456, FTE = 891

...

District O: PPE_instruction = \$11,234, FTE = 1,234

Numerator = $(12,345 \times 678) + (13,456 \times 891) + \dots + (11,234 \times 1,234)$

= 8,369,910 + 11,989,296 + ... + 13,862,756

= 156,432,890

Denominator = 678 + 891 + ... + 1,234 = 13,456 FTE

Weighted PPE_instruction = 156,432,890 / 13,456 = \$11,627.45/pupil

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4. SHADING THRESHOLD CALCULATIONS

Purpose: Determine 5% / 0.5pp thresholds for gradient shading in comparison tables

Source: Appendix A (Threshold Analysis), compose_pdf.py lines 1212-1349

Statistical Analysis (All 59 Western MA Districts):

1. Calculate variation in PPE:

- Mean PPE = \$24,237

- Std Dev PPE = \$5,462

- Coefficient of Variation (CV) = $5,462 / 24,237 = 0.225$ (22.5%)

2. Calculate variation in CAGR:

- Mean CAGR = 6.00 percentage points

- Std Dev CAGR = 3.24 percentage points

- Coefficient of Variation (CV) = $3.24 / 6.00 = 0.540$ (54.0%)

3. Key insight: CAGR varies 2.4× more than PPE relative to means ($54.0\% / 22.5\% = 2.4$)

4. Evaluate threshold scenarios:

Scenario: 5% PPE / 0.5pp CAGR (SELECTED)

- PPE: 5% = $0.05 \times 24,237 = \$1,212$

- PPE Standard Deviations: $1,212 / 5,462 = 0.22$ SD

- CAGR Standard Deviations: $0.50 / 3.24 = 0.15$ SD

- Balance ratio: $0.15 / 0.22 = 0.7\times$ (CAGR more sensitive for compound growth)

- PPE flagging rate: ~82% of comparisons

- CAGR flagging rate: ~88% of comparisons

Gradient Shading Bins:

- CAGR bins (absolute percentage points): [0.5pp, 1pp, 1.5pp, 2pp+]
- Dollar bins (relative percent): [5%, 10%, 15%, 20%+]
- Intensity increases with bin: lightest shade → darkest shade

5. DISTRICT COMPARISON TABLE CALCULATIONS

Purpose: Compare district PPE to baseline (cohort or Western MA aggregate)

Source: compose_pdf.py, lines 425-583 (_build_category_table)

For Each Category (e.g., Instruction, Student Support, etc.):

1. Get latest year (2024) and 5-year-ago baseline (2019)

2. Calculate CAGR:

District CAGR = $[(\text{PPE}_{2024} / \text{PPE}_{2019})^{(1/5)} - 1] \times 100$

Baseline CAGR = $[(\text{Baseline_PPE}_{2024} / \text{Baseline_PPE}_{2019})^{(1/5)} - 1] \times 100$

3. Calculate CAGR difference (absolute percentage points):

CAGR_diff = $|\text{District_CAGR} - \text{Baseline_CAGR}|$

4. Determine CAGR shading:

if CAGR_diff < 1.0pp: No shading (white)

elif CAGR_diff < 2.0pp: Lightest shade

elif CAGR_diff < 3.0pp: Light shade

elif CAGR_diff < 4.0pp: Medium shade

else: Darkest shade

Color: Amber if above baseline, Teal if below

5. Calculate dollar difference (relative percent):

Dollar_diff = $|\text{PPE}_{2024} - \text{Baseline_PPE}_{2024}| / \text{Baseline_PPE}_{2024}$

6. Determine dollar shading:

if Dollar_diff < 5%: No shading

elif Dollar_diff < 10%: Lightest shade

elif Dollar_diff < 15%: Light shade

elif Dollar_diff < 20%: Medium shade

else: Darkest shade

Color: Amber if above baseline, Teal if below

Example (Amherst Instruction vs Medium Cohort):

- Amherst PPE_2019 = \$10,234, PPE_2024 = \$12,456

- Medium PPE_2019 = \$9,876, PPE_2024 = \$11,234

CAGR calculations:

- Amherst CAGR = $[(12,456 / 10,234)^{0.2} - 1] \times 100 = 4.01\%$

- Medium CAGR = $[(11,234 / 9,876)^{0.2} - 1] \times 100 = 2.60\%$

- CAGR_diff = $|4.01 - 2.60| = 1.41\text{pp} \rightarrow$ Medium-dark amber shade (in [1pp, 1.5pp] bin)

Dollar calculations:

- Dollar_diff = $|12,456 - 11,234| / 11,234 = 10.88\%$
- 10.88% is in [10%, 15%) range → Light amber shade

6. ENROLLMENT FTE CALCULATIONS

Purpose: Calculate total FTE enrollment from component categories

Source: school_shared.py, prepare_district_epp_lines()

FTE Categories:

- Foundation Enrollment (low-income, ELL, special ed weighted)
- PK (Pre-Kindergarten)
- K-6 (Kindergarten through 6th grade)
- 7-12 (7th through 12th grade)

Calculation:

Total FTE(year) = Foundation(year) + PK(year) + K6(year) + 712(year)

Example (Amherst 2024):

- Foundation = 1,234.5 FTE
- PK = 89.2 FTE
- K-6 = 678.3 FTE
- 7-12 = 543.1 FTE
- Total FTE = $1,234.5 + 89.2 + 678.3 + 543.1 = 2,545.1$ FTE

7. NSS/CH70 FUNDING CALCULATIONS

Purpose: Calculate Chapter 70 aid and Net School Spending from components

Source: school_shared.py, prepare_district_nss_ch70()

Components:

- Foundation Budget: State-determined minimum spending level
- Required Local Contribution: Municipality's required contribution
- Chapter 70 Aid: State aid = Foundation - Required Local Contribution
- Actual Net School Spending (NSS): Actual spending by district

Calculations:

Ch70 Aid = Foundation Budget - Required Local Contribution

NSS Gap = Actual NSS - Foundation Budget

Total Spending = Required Local Contribution + Ch70 Aid + NSS Gap

Example (Amherst 2024):

- Foundation Budget = \$45,678,901
- Required Local Contribution = \$38,234,567
- Ch70 Aid = $\$45,678,901 - \$38,234,567 = \$7,444,334$
- Actual NSS = \$52,123,456
- NSS Gap = $\$52,123,456 - \$45,678,901 = \$6,444,555$

NOTES

- All dollar values shown are illustrative examples for demonstration purposes
 - Actual data values are stored in Appendix C (Data Tables)
 - All calculations use enrollment-weighted aggregations for cohort comparisons
 - CAGR calculations exclude negative or zero values
 - Shading applies independently to CAGR and dollar comparisons
 - Year-specific cohort assignments are used for all historical maps and scatterplots
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DETAILED WORKED EXAMPLES

The following sections provide complete worked examples for specific districts and cohorts:

This appendix provides calculation examples showing the source of numbers and demonstrating every step of the analysis for two examples:

1. Western MA Medium Cohort (aggregate cohort calculations)
2. Amherst-Pelham Regional (individual district calculations)

The following pages detail:

- Data sources and cohort determination methodology
- Weighted average per-pupil expenditure calculations
- Time series analysis and compound annual growth rate (CAGR) formulas
- Chapter 70 state aid and Net School Spending calculations
- Comparative analysis methods

PART 1: DATA SOURCES AND COHORT DETERMINATION

All calculations use data from the file "E2C_Hub_MA_DESE_Data.xlsx" which contains multiple sheets:

- Sheet "District Expend by Category": Contains expenditure and enrollment data with columns DIST_NAME, YEAR, IND_CAT, IND_SUBCAT, IND_VALUE, SOURCE
- Sheet "District Regions": Contains regional classifications and school types
- Sheet "profile_DataC70": Contains Chapter 70 and Net School Spending data

Data sources by sheet (per SOURCE column):

District Expend by Category - Expenditures per Pupil:

Source: Massachusetts Department of Elementary and Secondary Education (DESE)

Dataset: "District Expenditures by Spending Category"

Last updated: August 12, 2025

URL: <https://educationtocareer.data.mass.gov/Finance-and-Budget/District-Expenditures-by-Spending-Category/er3w-dyti/>

This dataset provides:

- Expenditures per pupil (EPP) by spending category for all Massachusetts school districts
- Annual data from FY1993 to present

- Eight spending categories: Teachers, Insurance/Retirement/Other, Pupil Services, Other Teaching Services, Operations and Maintenance, Instructional Leadership, Administration, Other

District Expend by Category - Student Enrollment:

Source: Massachusetts Department of Elementary and Secondary Education (DESE)

Dataset: "District Expenditures by Spending Category"

URL: <https://educationtocareer.data.mass.gov/Finance-and-Budget/District-Expenditures-by-Spending-Category/er3w-dyti/>

This dataset provides three enrollment indicators:

- In-District FTE Pupils: Students enrolled in district-operated schools (used for cohort calculations)
- Out-of-District FTE Pupils: District students placed in other schools
- Total FTE Pupils: Sum of In-District and Out-of-District FTE

profile_DataC70 - Chapter 70 and Net School Spending:

Source: Massachusetts Department of Elementary and Secondary Education (DESE)

Dataset: Chapter 70 District Profiles (compiled from various DESE sources)

URL: Various sources compiled in profile_DataC70 spreadsheet

This dataset provides:

- Chapter 70 Aid (c70aid): State aid received by each district
- Required Net School Spending (rqdnss2): Minimum spending required by state law
- Actual Net School Spending (actualNSS): Total district spending on education
- Foundation Enrollment (distfoundenro): District foundation enrollment used for Ch70 funding calculations

District Regions - Regional Classifications:

Source: DESE district profiles and regional service mappings

This dataset provides:

- EOHHS regional designations (Western MA, Central MA, etc.)
- School type classifications (Traditional, Regional, Charter, Virtual, etc.)

1.1 COHORT BOUNDARY DETERMINATION

Cohort boundaries are determined by reference to Interquartile Range (IQR) analysis on Fiscal Year 2024 Full-Time Equivalent (FTE) in-district enrollment for all traditional public school districts in Western Massachusetts. Cohort boundaries are rounded for easier readability.

Step 1: Extract FY24 In-district FTE enrollment data

From Sheet "District Expend by Category", filter rows where:

- IND_CAT = "Student Enrollment"
- IND_SUBCAT = "In-District FTE Pupils" (Full-Time Equivalent students enrolled in district-operated schools)
- YEAR = 2024

Step 2: Calculate quartile statistics

Using the 60 traditional districts in Western MA with valid data (including Springfield), calculate:

- First quartile (Q1) = 25th percentile = 154.7 In-district FTE
- Second quartile/Median (Q2) = 50th percentile = 792.0 In-district FTE
- Third quartile (Q3) = 75th percentile = 1,597.6 In-district FTE
- 90th percentile (P90) = 3,571.1 In-district FTE

Step 3: Define cohort boundaries

- TINY: 0 to 200 In-district FTE (from zero to Q1 rounded to nearest 100)
- SMALL: 201 to 800 In-district FTE (from max TINY +1 to Median rounded to nearest 100)
- MEDIUM: 801 to 1,600 In-district FTE (from max SMALL +1 to Q3 rounded to nearest 100)
- LARGE: 1,601 to 4,000 In-district FTE (from max MEDIUM +1 to P90 rounded to nearest 1,000)
- X-LARGE: 4,001 to 10K In-district FTE (from max LARGE +1 to fixed threshold of 10,000)
- OUTLIERS (Springfield): Greater than fixed threshold of 10,000 In-district FTE

Step 4: Assign each district to its cohort

For each district, compare its FY24 In-district FTE enrollment to the boundaries above.

Example: Amherst-Pelham Regional has 1,209.3 In-district FTE in FY24

Since $801 \leq 1,209.3 \leq 1,600$, Amherst-Pelham is assigned to the MEDIUM cohort.

Example: Gill-Montague Regional has 899.8 In-district FTE in FY24

Since $801 \leq 899.8 \leq 1,600$, Gill-Montague is assigned to the MEDIUM cohort.

Example: Belchertown has 2,067.8 In-district FTE in FY24

Since $1,601 \leq 2,067.8 \leq 4,000$, Belchertown is assigned to the LARGE cohort.

1.2 EXPENDITURE CATEGORIES

All Per-Pupil Expenditure (PPE) calculations use eight categories:

1. Teachers
2. Insurance, Retirement and Other
3. Pupil Services
4. Other Teaching Services
5. Operations and Maintenance
6. Instructional Leadership
7. Administration
8. Other

These categories are derived from the Massachusetts Department of Elementary and Secondary Education (DESE) function code groupings in end-of-year reporting, and provide a standardized breakdown of all district expenditures. Note: District budgeting practices typically use categories that do not match the DESE end-of-year reporting standard.

PART 2: WESTERN MA MEDIUM COHORT CALCULATIONS

The Medium cohort contains 15 districts with in-district enrollment between 801 and 1,600 FTE in FY24. This section demonstrates how weighted average Per-Pupil Expenditure (PPE) is calculated for the cohort.

2.1 MEMBER DISTRICTS

Districts in the Medium cohort (sorted by FY24 In-district FTE enrollment):

1. Mohawk Trail: 804.8 In-district FTE
2. Monson: 807.8 In-district FTE
3. Gill-Montague: 899.8 In-district FTE
4. Amherst: 1,001.5 In-district FTE
5. Hoosac Valley Regional: 1,002.7 In-district FTE

6. Palmer: 1,061.5 In-district FTE
7. Ware: 1,095.8 In-district FTE
8. Berkshire Hills: 1,174.4 In-district FTE
9. North Adams: 1,196.0 In-district FTE
10. Mount Greylock: 1,206.9 In-district FTE
11. Amherst-Pelham: 1,209.3 In-district FTE
12. Southwick-Tolland-Granville Regional School District: 1,312.8 In-district FTE
13. Greenfield: 1,394.0 In-district FTE
14. Easthampton: 1,403.7 In-district FTE
15. Central Berkshire: 1,585.9 In-district FTE

Total In-district FTE for Medium cohort = Sum of all member In-district FTE = 17,156.9 FTE

2.2 WEIGHTED AVERAGE PPE CALCULATION (EXAMPLE: FY2024 TEACHERS)

For each expenditure category, we calculate an enrollment-weighted average across all districts in the cohort.

Step 1: Extract per-pupil expenditure data for Teachers category

From Sheet "District Expend by Category", filter rows where:

- IND_CAT = "Expenditures per Pupil"
- IND_SUBCAT = "Teachers"
- YEAR = 2024

Step 2: For each district, multiply PPE by enrollment weight

Example calculations for first three districts:

Mohawk Trail:

- FY24 Teachers PPE = [actual value from data] per pupil
- FY24 In-district FTE Enrollment = 804.8 FTE
- Weight = $804.8 / 17,156.9 = 0.0469$ (4.69% of cohort enrollment)
- Weighted contribution = [Teachers PPE] $\times$ 0.0469

Monson:

- FY24 Teachers PPE = [actual value from data] per pupil
- FY24 In-district FTE Enrollment = 807.8 FTE
- Weight = $807.8 / 17,156.9 = 0.0471$ (4.71% of cohort enrollment)
- Weighted contribution = [Teachers PPE] $\times$ 0.0471

Gill-Montague:

- FY24 Teachers PPE = [actual value from data] per pupil
- FY24 In-district FTE Enrollment = 899.8 FTE
- Weight = $899.8 / 17,156.9 = 0.0524$ (5.24% of cohort enrollment)
- Weighted contribution = [Teachers PPE] $\times$ 0.0524

[Continue for all 15 districts...]

Step 3: Sum all weighted contributions

Weighted Average Teachers PPE = Sum of all weighted contributions = [calculated value] per pupil

This weighted average calculation is repeated for all eight categories:

1. Teachers
2. Insurance, Retirement and Other
3. Pupil Services
4. Other Teaching Services
5. Operations and Maintenance
6. Instructional Leadership
7. Administration
8. Other

Step 4: Calculate total weighted average PPE

Total Weighted Average PPE (FY24) = Sum of all eight category weighted averages

= Teachers + Insurance + Pupil Services + Other Teaching + Operations + Instructional Leadership + Administration + Other

= [sum of actual calculated values] per pupil

2.3 TIME SERIES CALCULATIONS

The same weighted average calculation is performed for each fiscal year from 2009 to 2024, producing a time series of cohort-level PPE values.

For each year Y from 2009 to 2024:

1. Extract FY Y enrollment for all 13 Medium cohort districts
2. Calculate cohort total enrollment for year Y
- 3. For each of the 8 expenditure categories:**
 - a. Extract PPE values for year Y
 - b. Calculate enrollment weight for each district in year Y
 - c. Calculate weighted average PPE for that category in year Y
4. Sum the 8 category weighted averages to get total PPE for year Y

This produces arrays:

- Years: [2009, 2010, 2011, ..., 2024]
- Total PPE: [PPE_2009, PPE_2010, PPE_2011, ..., PPE_2024]
- Teachers PPE: [Teachers_2009, Teachers_2010, ..., Teachers_2024]
- [Similar arrays for other 7 categories]

2.4 COMPOUND ANNUAL GROWTH RATE (CAGR)

CAGR measures the annualized growth rate over multiple years.

Formula:

$$\text{CAGR} = ((\text{Value_end} / \text{Value_start})^{(1/\text{number_of_years})} - 1) \times 100\%$$

Where:

- Value_end = PPE in final year (2024)
- Value_start = PPE in initial year (2009)
- number_of_years = 2024 - 2009 = 15 years

Example for Medium cohort Total PPE:

- PPE_2009 = [calculated weighted average for 2009] per pupil

- PPE_2024 = [calculated weighted average for 2024] per pupil
- Number of years = 15

$$\text{CAGR} = ((\text{PPE}_{2024} / \text{PPE}_{2009})^{(1/15)} - 1) \times 100\%$$

 = [calculated value] % per year

This calculation is performed for:

- Total PPE CAGR
- Each of the 8 category-specific PPE CAGRs
- Chapter 70 state aid CAGR
- Net School Spending (NSS) CAGR
- Enrollment CAGR

2.5 YEAR-OVER-YEAR (YOY) GROWTH RATE

Year-over-Year (YoY) growth rate measures the percentage change from one year to the next, providing insight into annual fluctuations in expenditures.

Formula:

$$\text{YoY Growth Rate} = ((\text{Value}_{\text{year}} / \text{Value}_{\text{previous_year}}) - 1) \times 100\%$$

Where:

- Value_year = PPE in current year
- Value_previous_year = PPE in previous year

Example for Medium cohort Total PPE:

Assume the time series shows:

- PPE_2009 = \$12,000 per pupil
- PPE_2010 = \$12,500 per pupil
- PPE_2011 = \$13,000 per pupil
- PPE_2012 = \$13,100 per pupil
- ...

YoY calculations:

- $\text{YoY}_{2010} = ((\$12,500 / \$12,000) - 1) \times 100\% = 4.17\%$
- $\text{YoY}_{2011} = ((\$13,000 / \$12,500) - 1) \times 100\% = 4.00\%$
- $\text{YoY}_{2012} = ((\$13,100 / \$13,000) - 1) \times 100\% = 0.77\%$
- ...

Key differences from CAGR:

- YoY shows year-to-year changes; CAGR smooths growth over multiple years
- YoY captures annual volatility; CAGR assumes constant growth
- YoY is useful for identifying specific years with unusual growth patterns

Example comparison:

If CAGR from 2009 to 2024 is 3.71% per year, but YoY rates vary from 0.5% to 8.0% across individual years, this reveals that growth was not constant—some years had much higher or lower increases than the average.

Use cases:

- Identifying years with budget constraints (low YoY)

- Identifying years with significant increases (high YoY)
- Understanding the impact of specific events (e.g., pandemic, policy changes) on year-to-year spending

This YoY calculation is performed for:

- Total PPE YoY for each year from 2010 to 2024 (15 values)
- Each of the 8 category-specific PPE YoY rates
- Chapter 70 state aid YoY
- Net School Spending (NSS) YoY
- Enrollment YoY

Note: YoY for year N requires data from year N-1, so the first YoY value is for 2010 (comparing 2010 to 2009).

PART 3: AMHERST-PELHAM REGIONAL DISTRICT CALCULATIONS

This section demonstrates all calculations for a single district: Amherst-Pelham Regional.

3.1 DISTRICT PROFILE

District: Amherst-Pelham Regional

Cohort Assignment: MEDIUM (801-1,600 In-district FTE)

FY24 In-district FTE Enrollment: 1,209.3 FTE

Data Source: Appendix A Data Tables

- Enrollment data: See "In-District FTE" column in Appendix A
- PPE data: See expenditure category columns in Appendix A
- Chapter 70 data: See "Chapter 70 Aid" column in Appendix A
- NSS data: See "Required NSS" and "Actual NSS" columns in Appendix A

3.2 ENROLLMENT TIME SERIES

Step 1: Extract in-district enrollment data

From Appendix A Data Tables, locate:

- District: "Amherst-Pelham Regional"
- Find the "In-District FTE" column
- Read values for all years from 2009 to 2024

Step 2: Create time series array

Years: [2009, 2010, 2011, ..., 2023, 2024]

In-district FTE: [FTE_2009, FTE_2010, FTE_2011, ..., FTE_2023, 1,209.3]

(Extract actual values for each year from Appendix A)

Step 3: Calculate enrollment CAGR

$$\text{CAGR} = ((\text{FTE_2024} / \text{FTE_2009})^{(1/15)} - 1) \times 100\%$$

$$= ((1,209.3 / \text{FTE_2009})^{(1/15)} - 1) \times 100\%$$

$$= [\text{calculated value}] \% \text{ per year}$$

Where:

- FTE_2024 = 1,209.3 In-district FTE (from data)

- $FTE_{2009} = [\text{actual value from data}] \text{ In-district FTE}$
- Number of years = 15 (2024 - 2009)

A negative CAGR indicates declining enrollment, positive indicates growth.

3.3 PER-PUPIL EXPENDITURE CALCULATIONS

For Amherst-Pelham Regional, PPE is calculated for each of the eight categories:

1. Teachers
2. Insurance, Retirement and Other
3. Pupil Services
4. Other Teaching Services
5. Operations and Maintenance
6. Instructional Leadership
7. Administration
8. Other

Step 1: Extract PPE data for each category

From Appendix A Data Tables, locate:

- District: "Amherst-Pelham Regional"
- Find the columns for each of the 8 expenditure categories
- Read values for all years from 2009 to 2024

Step 2: Example calculation for Teachers category (FY2024)

From Appendix A Data Tables:

- District: "Amherst-Pelham Regional"
- Year: 2024
- Column: "Teachers"
- Value: [actual value from Appendix A] per pupil

This value represents the amount Amherst-Pelham Regional spent per Full-Time Equivalent (FTE) student on Teachers in FY2024.

Step 3: Calculate total PPE for FY2024

Total PPE (FY2024) = Teachers + Insurance, Retirement and Other + Pupil Services + Other Teaching Services + Operations and Maintenance + Instructional Leadership + Administration + Other

Using column IND_VALUE for YEAR = 2024 and each category:

- Teachers: [actual value]
- Insurance, Retirement and Other: [actual value]
- Pupil Services: [actual value]
- Other Teaching Services: [actual value]
- Operations and Maintenance: [actual value]
- Instructional Leadership: [actual value]
- Administration: [actual value]
- Other: [actual value]

Total PPE (FY2024) = [sum of actual values] per pupil

Step 4: Repeat for all years 2009-2024

For each year from 2009 to 2024, extract IND_VALUE for each of the 8 categories and sum them to get total PPE for that year.

This produces time series arrays:

- Years: [2009, 2010, ..., 2024]
- Total PPE: [PPE_2009, PPE_2010, ..., PPE_2024]
- Teachers: [Teachers_2009, Teachers_2010, ..., Teachers_2024]
- Insurance, Retirement and Other: [Insurance_2009, Insurance_2010, ..., Insurance_2024]
- Pupil Services: [Pupil_Services_2009, Pupil_Services_2010, ..., Pupil_Services_2024]
- Other Teaching Services: [Other_Teaching_2009, Other_Teaching_2010, ..., Other_Teaching_2024]
- Operations and Maintenance: [Operations_2009, Operations_2010, ..., Operations_2024]
- Instructional Leadership: [Instructional_Leadership_2009, Instructional_Leadership_2010, ..., Instructional_Leadership_2024]
- Administration: [Administration_2009, Administration_2010, ..., Administration_2024]
- Other: [Other_2009, Other_2010, ..., Other_2024]

Step 5: Calculate CAGR for each category

Formula: $CAGR = ((Value_{2024} / Value_{2009})^{(1/15)} - 1) \times 100\%$

Example for Teachers:

- Teachers_2009 = [actual value from data] per pupil
- Teachers_2024 = [actual value from data] per pupil
- $CAGR_Teachers = ((Teachers_{2024} / Teachers_{2009})^{(1/15)} - 1) \times 100\%$
= [calculated value] % per year

Repeat this calculation for all 8 categories plus total PPE to get 9 CAGR values.

3.4 CHAPTER 70 STATE AID CALCULATIONS

Chapter 70 is the major state aid program for Massachusetts K-12 public schools.

Step 1: Extract Chapter 70 data

From Appendix A Data Tables, locate:

- District: "Amherst-Pelham Regional"
- Column: "Chapter 70 Aid"
- Read values for all years from 2009 to 2024

Step 2: Extract total district expenditures

From Appendix A Data Tables, calculate total expenditure as:

- District: "Amherst-Pelham Regional"
- Sum of all expenditure categories for each year
- Alternatively, use "Actual NSS" column if available

Step 3: Calculate Chapter 70 as percentage of total expenditure (FY2024 example)

- Chapter 70 Aid (FY2024) = [value from "Chapter 70 Aid" column in Appendix A for 2024]
- Total Expenditure (FY2024) = [sum of expenditure categories from Appendix A for 2024]
- Percentage = $(Chapter\ 70\ Aid / Total\ Expenditure) \times 100\%$

This percentage shows what portion of the district's total expenditure came from Chapter 70 state aid in FY2024.

Step 4: Calculate Chapter 70 per-pupil (FY2024 example)

- Chapter 70 Aid (FY2024) = [value from "Chapter 70 Aid" column in Appendix A]
- Enrollment (FY2024) = 1,209.3 In-district FTE
- Chapter 70 Per-Pupil = Chapter 70 Aid / 1,209.3 FTE = [calculated value] per FTE

Step 5: Calculate Chapter 70 CAGR

- Chapter 70 Aid (2009) = [value from "Chapter 70 Aid" column in Appendix A for 2009]
- Chapter 70 Aid (2024) = [value from "Chapter 70 Aid" column in Appendix A for 2024]
- CAGR = $((\text{Ch70Aid_2024} / \text{Ch70Aid_2009})^{(1/15)} - 1) \times 100\%$
= [calculated value] % per year

A negative CAGR indicates that Chapter 70 aid decreased, positive indicates growth.

3.5 NET SCHOOL SPENDING (NSS) CALCULATIONS

Net School Spending (NSS) is the total amount a school district spends on education from all sources, excluding certain items like transportation, debt service, and capital expenditures.

Step 1: Extract NSS data

From Appendix A Data Tables, locate:

- District: "Amherst-Pelham Regional"
- Columns: "Required NSS" and "Actual NSS"
- Read values for all years from 2009 to 2024

Step 2: Calculate NSS per-pupil (FY2024 example)

- Total NSS (FY2024) = [value from "Actual NSS" column in Appendix A for 2024]
- Enrollment (FY2024) = 1,209.3 In-district FTE
- NSS Per-Pupil = Total NSS / 1,209.3 FTE = [calculated value] per FTE

Step 3: Calculate NSS CAGR

- NSS (2009) = [value from "Actual NSS" column in Appendix A for 2009]
- NSS (2024) = [value from "Actual NSS" column in Appendix A for 2024]
- CAGR = $((\text{NSS_2024} / \text{NSS_2009})^{(1/15)} - 1) \times 100\%$
= [calculated value] % per year

This indicates the average annual growth rate of NSS from 2009 to 2024.

Step 4: Compare NSS per-pupil to cohort average

Using the weighted average calculation method described in Part 2, the MEDIUM cohort average NSS per-pupil (FY2024) = [calculated weighted average] per FTE.

Amherst-Pelham's NSS per-pupil ([actual value from calculation]) is [percentage difference]% [above/below] the MEDIUM cohort average.

Calculation: $((\text{Amherst-Pelham NSS per-pupil} - \text{MEDIUM cohort average}) / \text{MEDIUM cohort average}) \times 100\%$

3.6 COMPARATIVE ANALYSIS

All district-specific calculations can be compared to:

1. Cohort averages (in this case, MEDIUM cohort weighted averages for Amherst-Pelham Regional)
2. Regional averages (all Western MA traditional districts)

3. Statewide averages (if available)

For each metric (Total PPE, each of 8 category PPEs, Chapter 70 per-pupil, NSS per-pupil, enrollment growth), calculate:

- District value
- Cohort average
- Difference = District value - Cohort average
- Percentage difference = $(\text{Difference} / \text{Cohort average}) \times 100\%$

These comparisons reveal whether a district is spending more or less than similar-sized districts, and where specific spending differences occur across the eight categories:

1. Teachers
2. Insurance, Retirement and Other
3. Pupil Services
4. Other Teaching Services
5. Operations and Maintenance
6. Instructional Leadership
7. Administration
8. Other

Appendix C. Data Tables

All data values used in plots

This appendix contains the underlying data tables for all districts and regions shown in the report. Each table shows PPE by category (in \$/pupil), FTE enrollment counts, and NSS/Ch70 funding components (in absolute dollars) across all available years.

Data: All Western MA Traditional Districts: Tiny (0-200 FTE)

PPE (\$/pupil), FTE Enrollment, and NSS/Ch70 Funding (\$)

PPE by Category (\$/pupil)

Table 1																
Category	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Other	\$577	\$571	\$623	\$643	\$647	\$665	\$676	\$709	\$780	\$579	\$673	\$738	\$1,380	\$1,042	\$1,304	\$1,057
Guidance, Counseling and Testing	\$345	\$365	\$375	\$402	\$419	\$462	\$475	\$454	\$445	\$448	\$459	\$488	\$577	\$663	\$621	\$674
Administration	\$770	\$808	\$814	\$853	\$915	\$851	\$774	\$819	\$863	\$869	\$770	\$833	\$950	\$1,003	\$1,114	\$1,126
Instructional Leadership	\$1,042	\$1,096	\$1,125	\$1,225	\$1,269	\$1,316	\$1,496	\$1,584	\$1,510	\$1,524	\$1,543	\$1,589	\$1,817	\$1,733	\$1,777	\$1,991
Operations and Maintenance	\$1,374	\$1,312	\$1,460	\$1,416	\$1,359	\$1,558	\$1,554	\$1,452	\$1,433	\$1,468	\$1,811	\$1,670	\$2,060	\$2,389	\$2,232	\$2,652
Other Teaching Services	\$1,899	\$1,926	\$1,987	\$2,112	\$2,412	\$2,378	\$2,494	\$2,635	\$2,673	\$2,707	\$3,004	\$3,115	\$3,322	\$3,677	\$3,638	\$3,757
Pupil Services	\$1,324	\$1,427	\$1,426	\$1,497	\$1,605	\$1,599	\$1,670	\$1,753	\$1,846	\$1,813	\$1,864	\$1,716	\$2,119	\$2,010	\$2,334	\$2,732
Insurance, Retirement and Other	\$2,258	\$2,405	\$2,595	\$2,737	\$2,926	\$3,148	\$3,125	\$3,143	\$3,905	\$3,390	\$3,662	\$3,769	\$4,357	\$4,048	\$4,506	\$5,009
Teachers	\$5,148	\$5,382	\$5,545	\$5,277	\$5,430	\$5,906	\$6,187	\$6,222	\$6,245	\$6,520	\$6,792	\$6,982	\$8,015	\$7,858	\$7,808	\$8,507
Total	\$14,738	\$15,291	\$15,950	\$16,161	\$16,981	\$17,882	\$18,452	\$18,770	\$19,700	\$19,319	\$20,577	\$20,900	\$24,596	\$24,423	\$25,334	\$27,504

FTE Enrollment

Table 2																
FTE Series	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
In-District FTE Pupils	131	127	125	125	123	120	122	116	121	123	122	122	107	115	114	113
Foundation Enrollment	123	120	116	109	109	108	107	111	108	107	105	102	101	91	93	94
Out-of-District FTE Pupils	41	41	41	37	40	38	37	39	38	36	35	33	31	34	34	32

NSS/Ch70 Funding Components (\$)

Table 3																
Component	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Ch70 Aid	\$3K	\$3K	\$3K	\$4K	\$4K	\$4K	\$4K	\$4K	\$4K	\$4K	\$4K	\$4K	\$4K	\$5K	\$5K	\$5K
Req NSS (minus Ch70)	\$6K	\$6K	\$6K	\$7K	\$7K	\$7K	\$7K	\$8K	\$8K	\$8K	\$8K	\$8K	\$9K	\$9K	\$9K	\$10K
Actual NSS (minus Req NSS)	\$5K	\$4K	\$5K	\$5K	\$6K	\$6K	\$6K	\$6K	\$8K	\$8K	\$10K	\$10K	\$10K	\$12K	\$12K	\$12K
Total NSS	\$13K	\$14K	\$15K	\$16K	\$17K	\$17K	\$17K	\$18K	\$20K	\$20K	\$22K	\$22K	\$23K	\$26K	\$26K	\$27K

Data: All Western MA Traditional Districts: Small (201-800 FTE)

PPE (\$/pupil), FTE Enrollment, and NSS/Ch70 Funding (\$)

PPE by Category (\$/pupil)

Table 4

Category	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Other	\$558	\$505	\$540	\$533	\$585	\$515	\$503	\$583	\$539	\$596	\$558	\$512	\$915	\$777	\$1,064	\$1,150
Guidance, Counseling and Testing	\$392	\$422	\$418	\$456	\$483	\$482	\$501	\$564	\$588	\$600	\$669	\$662	\$807	\$864	\$896	\$961
Administration	\$523	\$515	\$567	\$601	\$648	\$665	\$701	\$739	\$999	\$752	\$788	\$802	\$854	\$942	\$948	\$976
Instructional Leadership	\$811	\$805	\$838	\$824	\$909	\$947	\$967	\$1,024	\$1,028	\$1,042	\$1,016	\$1,097	\$1,188	\$1,260	\$1,336	\$1,463
Operations and Maintenance	\$1,108	\$1,056	\$1,043	\$1,101	\$1,198	\$1,264	\$1,295	\$1,338	\$1,254	\$1,285	\$1,473	\$1,422	\$1,676	\$1,878	\$2,176	\$2,454
Other Teaching Services	\$1,061	\$1,089	\$1,193	\$1,151	\$1,289	\$1,311	\$1,353	\$1,446	\$1,575	\$1,718	\$1,799	\$1,806	\$1,779	\$1,899	\$2,151	\$2,402
Pupil Services	\$1,421	\$1,366	\$1,441	\$1,537	\$1,733	\$1,739	\$1,714	\$1,913	\$1,850	\$1,870	\$1,974	\$2,007	\$2,097	\$2,569	\$2,760	\$3,122
Insurance, Retirement and Other	\$2,128	\$2,107	\$2,230	\$2,317	\$2,486	\$2,568	\$2,711	\$2,955	\$3,219	\$3,345	\$3,362	\$3,420	\$3,734	\$3,801	\$3,969	\$4,254
Teachers	\$4,522	\$4,588	\$4,679	\$4,870	\$5,353	\$5,423	\$5,557	\$5,771	\$6,026	\$6,075	\$6,152	\$6,450	\$6,922	\$7,338	\$7,596	\$7,838
Total	\$12,523	\$12,453	\$12,949	\$13,388	\$14,684	\$14,916	\$15,302	\$16,333	\$17,077	\$17,283	\$17,791	\$18,177	\$19,972	\$21,330	\$22,896	\$24,622

FTE Enrollment

Table 5

FTE Series	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
In-District FTE Pupils	775	779	763	742	696	680	661	644	628	614	617	601	560	556	553	542
Foundation Enrollment	695	676	666	646	628	603	590	559	542	536	528	526	522	494	484	481
Out-of-District FTE Pupils	75	78	73	73	80	82	86	82	85	88	90	88	85	87	87	84

NSS/Ch70 Funding Components (\$)

Table 6

Component	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Ch70 Aid	\$4K	\$4K	\$4K	\$4K	\$4K	\$4K	\$4K	\$5K	\$5K	\$5K	\$5K	\$5K	\$5K	\$6K	\$6K	\$6K
Req NSS (minus Ch70)	\$5K	\$6K	\$6K	\$6K	\$6K	\$7K	\$7K	\$7K	\$7K	\$7K	\$7K	\$7K	\$8K	\$8K	\$9K	\$9K
Actual NSS (minus Req NSS)	\$2K	\$2K	\$2K	\$2K	\$2K	\$2K	\$3K	\$4K	\$4K	\$4K	\$5K	\$5K	\$5K	\$5K	\$6K	\$6K
Total NSS	\$11K	\$11K	\$11K	\$12K	\$13K	\$14K	\$14K	\$15K	\$16K	\$16K	\$17K	\$17K	\$18K	\$19K	\$21K	\$21K

Data: All Western MA Traditional Districts: Medium (801-1600 FTE)

PPE (\$/pupil), FTE Enrollment, and NSS/Ch70 Funding (\$)

PPE by Category (\$/pupil)

Table 7

Category	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Other	\$461	\$484	\$558	\$494	\$561	\$552	\$524	\$573	\$586	\$534	\$644	\$607	\$930	\$1,064	\$1,055	\$1,200
Guidance, Counseling and Testing	\$373	\$385	\$391	\$418	\$435	\$457	\$470	\$479	\$522	\$536	\$545	\$601	\$677	\$736	\$805	\$826
Administration	\$494	\$521	\$512	\$533	\$575	\$576	\$614	\$653	\$678	\$711	\$653	\$700	\$747	\$785	\$835	\$911
Instructional Leadership	\$893	\$902	\$885	\$1,039	\$1,062	\$1,091	\$1,113	\$1,169	\$1,150	\$1,071	\$1,063	\$1,113	\$1,191	\$1,263	\$1,373	\$1,573
Operations and Maintenance	\$1,150	\$1,085	\$1,155	\$1,095	\$1,146	\$1,208	\$1,233	\$1,218	\$1,228	\$1,203	\$1,326	\$1,383	\$1,564	\$1,663	\$1,836	\$1,953
Other Teaching Services	\$1,014	\$1,078	\$1,100	\$1,135	\$1,235	\$1,317	\$1,362	\$1,400	\$1,444	\$1,596	\$1,681	\$1,729	\$1,936	\$2,180	\$2,311	\$2,395
Pupil Services	\$1,351	\$1,309	\$1,366	\$1,345	\$1,466	\$1,527	\$1,626	\$1,705	\$1,686	\$1,869	\$1,919	\$1,803	\$1,957	\$2,231	\$2,548	\$2,755
Insurance, Retirement and Other	\$2,483	\$2,574	\$2,743	\$2,707	\$2,707	\$2,943	\$3,074	\$3,251	\$3,266	\$3,622	\$3,631	\$3,789	\$4,169	\$4,166	\$4,480	\$4,740
Teachers	\$4,813	\$4,890	\$5,055	\$5,048	\$5,152	\$5,357	\$5,513	\$5,578	\$5,674	\$5,960	\$6,251	\$6,385	\$6,758	\$6,773	\$7,182	\$7,431
Total	\$13,032	\$13,227	\$13,766	\$13,815	\$14,340	\$15,028	\$15,529	\$16,025	\$16,234	\$17,103	\$17,713	\$18,110	\$19,930	\$20,859	\$22,426	\$23,785

FTE Enrollment

Table 8

FTE Series	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
In-District FTE Pupils	1,466	1,434	1,416	1,402	1,388	1,334	1,306	1,277	1,268	1,247	1,268	1,249	1,160	1,175	1,158	1,144
Foundation Enrollment	1,622	1,596	1,567	1,528	1,509	1,507	1,472	1,451	1,422	1,399	1,381	1,376	1,361	1,289	1,283	1,247
Out-of-District FTE Pupils	154	161	164	164	166	167	176	185	187	194	201	198	190	191	194	194

NSS/Ch70 Funding Components (\$)

Table 9

Component	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Ch70 Aid	\$5K	\$5K	\$5K	\$5K	\$5K	\$6K	\$6K	\$6K	\$6K	\$6K	\$6K	\$7K	\$7K	\$7K	\$8K	\$8K
Req NSS (minus Ch70)	\$4K	\$5K	\$5K	\$5K	\$5K	\$5K	\$6K	\$6K	\$6K	\$6K	\$6K	\$6K	\$6K	\$7K	\$7K	\$8K
Actual NSS (minus Req NSS)	\$2K	\$1K	\$2K	\$2K	\$2K	\$2K	\$2K	\$2K	\$2K	\$3K	\$3K	\$3K	\$3K	\$3K	\$4K	\$4K
Total NSS	\$11K	\$11K	\$11K	\$12K	\$12K	\$13K	\$13K	\$14K	\$14K	\$15K	\$16K	\$16K	\$16K	\$17K	\$18K	\$20K

Data: All Western MA Traditional Districts: Large (1601-10K FTE)

PPE (\$/pupil), FTE Enrollment, and NSS/Ch70 Funding (\$)

PPE by Category (\$/pupil)

Table 10

Category	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Other	\$513	\$648	\$614	\$611	\$654	\$631	\$610	\$644	\$664	\$667	\$722	\$716	\$972	\$1,075	\$1,331	\$1,348
Guidance, Counseling and Testing	\$397	\$394	\$416	\$407	\$423	\$443	\$452	\$454	\$454	\$483	\$511	\$537	\$597	\$664	\$741	\$740
Administration	\$392	\$415	\$426	\$437	\$441	\$462	\$484	\$496	\$480	\$494	\$453	\$502	\$537	\$571	\$616	\$649
Instructional Leadership	\$811	\$859	\$865	\$856	\$837	\$886	\$915	\$932	\$956	\$940	\$998	\$1,028	\$1,060	\$1,153	\$1,195	\$1,231
Operations and Maintenance	\$1,014	\$993	\$1,009	\$986	\$984	\$1,023	\$1,093	\$1,098	\$1,111	\$1,152	\$1,244	\$1,238	\$1,401	\$1,450	\$1,685	\$1,683
Other Teaching Services	\$990	\$1,037	\$1,103	\$1,105	\$1,177	\$1,228	\$1,290	\$1,365	\$1,383	\$1,395	\$1,407	\$1,429	\$1,561	\$1,717	\$1,951	\$2,021
Pupil Services	\$1,135	\$1,141	\$1,206	\$1,205	\$1,274	\$1,354	\$1,396	\$1,472	\$1,495	\$1,586	\$1,659	\$1,591	\$1,603	\$1,989	\$2,213	\$2,408
Insurance, Retirement and Other	\$1,911	\$1,943	\$2,045	\$2,123	\$2,115	\$2,282	\$2,366	\$2,419	\$2,543	\$2,652	\$2,773	\$2,855	\$3,037	\$3,212	\$3,364	\$3,469
Teachers	\$4,613	\$4,604	\$4,707	\$4,868	\$4,990	\$5,067	\$5,194	\$5,393	\$5,482	\$5,569	\$5,863	\$6,031	\$6,461	\$6,780	\$7,001	\$7,292
Total	\$11,775	\$12,034	\$12,391	\$12,597	\$12,895	\$13,376	\$13,801	\$14,273	\$14,568	\$14,937	\$15,629	\$15,927	\$17,229	\$18,612	\$20,097	\$20,841

FTE Enrollment

Table 11

FTE Series	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
In-District FTE Pupils	4,001	3,953	3,935	3,852	3,812	3,760	3,721	3,674	3,690	3,680	3,616	3,598	3,397	3,422	3,384	3,379
Foundation Enrollment	4,432	4,382	4,347	4,325	4,280	4,230	4,191	4,163	4,093	4,077	4,037	4,013	3,975	3,823	3,787	3,752
Out-of-District FTE Pupils	215	227	244	253	256	272	293	309	316	322	341	345	344	341	354	357

NSS/Ch70 Funding Components (\$)

Table 12

Component	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Ch70 Aid	\$6K	\$5K	\$6K	\$6K	\$6K	\$6K	\$6K	\$6K	\$6K	\$7K	\$7K	\$7K	\$7K	\$8K	\$9K	\$10K
Req NSS (minus Ch70)	\$3K	\$4K	\$4K	\$4K	\$4K	\$4K	\$5K	\$5K	\$5K	\$5K	\$5K	\$5K	\$5K	\$5K	\$6K	\$6K
Actual NSS (minus Req NSS)	\$900	\$708	\$901	\$905	\$944	\$1K	\$1K	\$1K	\$2K	\$2K	\$2K	\$2K	\$2K	\$2K	\$2K	\$2K
Total NSS	\$10K	\$10K	\$10K	\$11K	\$11K	\$12K	\$12K	\$12K	\$13K	\$13K	\$14K	\$14K	\$15K	\$16K	\$17K	\$18K

Data: All Western MA Traditional Districts: Outliers (Springfield at 23,723 FTE in 2024)

PPE (\$/pupil), FTE Enrollment, and NSS/Ch70 Funding (\$)

PPE by Category (\$/pupil)

Table 13																
Category	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Other	\$1,143	\$1,337	\$2,068	\$1,143	\$1,424	\$1,734	\$1,241	\$1,376	\$1,286	\$1,371	\$1,673	\$1,765	\$2,487	\$2,260	\$2,923	\$3,071
Guidance, Counseling and Testing	\$380	\$362	\$427	\$388	\$400	\$405	\$408	\$389	\$391	\$438	\$469	\$483	\$513	\$522	\$621	\$697
Administration	\$533	\$654	\$446	\$645	\$550	\$677	\$596	\$580	\$507	\$490	\$416	\$407	\$441	\$501	\$599	\$609
Instructional Leadership	\$719	\$778	\$892	\$1,173	\$1,257	\$1,245	\$1,208	\$1,282	\$1,280	\$1,060	\$983	\$1,090	\$1,103	\$1,402	\$1,534	\$1,571
Operations and Maintenance	\$1,150	\$1,081	\$1,262	\$1,176	\$1,132	\$997	\$1,121	\$1,027	\$1,063	\$1,156	\$1,337	\$1,093	\$1,534	\$1,536	\$3,453	\$3,806
Other Teaching Services	\$1,143	\$1,002	\$1,146	\$1,130	\$1,051	\$1,086	\$1,150	\$1,135	\$1,070	\$1,036	\$1,357	\$1,390	\$1,337	\$1,518	\$1,776	\$2,014
Pupil Services	\$1,651	\$1,511	\$1,729	\$1,965	\$1,783	\$1,852	\$1,985	\$2,078	\$2,222	\$2,407	\$2,499	\$2,148	\$2,077	\$3,080	\$3,054	\$3,204
Insurance, Retirement and Other	\$2,050	\$2,226	\$2,280	\$2,474	\$2,437	\$2,496	\$2,467	\$2,566	\$2,562	\$2,635	\$2,765	\$3,001	\$3,217	\$3,461	\$3,852	\$3,906
Teachers	\$5,839	\$4,859	\$4,750	\$4,711	\$4,848	\$4,823	\$4,963	\$5,196	\$4,962	\$5,274	\$5,510	\$5,800	\$6,442	\$7,095	\$7,561	\$7,737
Total	\$14,608	\$13,810	\$15,000	\$14,805	\$14,882	\$15,315	\$15,139	\$15,629	\$15,343	\$15,867	\$17,009	\$17,177	\$19,151	\$21,375	\$25,373	\$26,615

FTE Enrollment

Table 14																
FTE Series	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
In-District FTE Pupils	25,137	24,987	24,935	24,521	24,747	25,249	25,101	25,009	25,629	25,776	25,294	24,906	24,079	24,040	23,763	23,723
Foundation Enrollment	28,235	28,343	28,305	28,226	27,951	28,434	28,755	28,970	29,109	29,533	29,670	29,645	29,551	29,144	28,613	28,326
Out-of-District FTE Pupils	3,571	3,775	3,719	3,695	3,817	3,613	3,974	4,406	4,596	4,812	5,150	5,363	5,559	5,500	5,652	5,745

NSS/Ch70 Funding Components (\$)

Table 15																
Component	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Ch70 Aid	\$9K	\$9K	\$9K	\$10K	\$10K	\$10K	\$10K	\$11K	\$11K	\$11K	\$12K	\$13K	\$13K	\$14K	\$15K	\$17K
Req NSS (minus Ch70)	\$247	\$1K	\$2K	\$1K	\$1K	\$1K	\$1K	\$1K	\$1K	\$1K	\$1K	\$1K	\$2K	\$1K	\$2K	\$2K
Actual NSS (minus Req NSS)	\$-150	\$-384	\$-35	\$-84	\$-144	\$0	\$2	\$2	\$-0	\$-106	\$-0	\$-499	\$0	\$0	\$-0	\$-0
Total NSS	\$9K	\$10K	\$11K	\$11K	\$11K	\$12K	\$12K	\$12K	\$12K	\$12K	\$13K	\$13K	\$15K	\$15K	\$17K	\$19K

Data: All Western MA Traditional Districts: All Western MA (excl. Springfield)

PPE (\$/pupil), FTE Enrollment, and NSS/Ch70 Funding (\$)

PPE by Category (\$/pupil)

Table 16

Category	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Other	\$507	\$590	\$593	\$575	\$624	\$600	\$580	\$622	\$636	\$627	\$686	\$671	\$967	\$1,043	\$1,240	\$1,287
Guidance, Counseling and Testing	\$389	\$394	\$409	\$415	\$432	\$451	\$462	\$471	\$483	\$506	\$534	\$564	\$636	\$700	\$768	\$780
Administration	\$440	\$461	\$471	\$488	\$507	\$520	\$544	\$566	\$589	\$580	\$542	\$587	\$628	\$669	\$713	\$754
Instructional Leadership	\$836	\$869	\$873	\$906	\$910	\$951	\$981	\$1,013	\$1,023	\$996	\$1,029	\$1,070	\$1,122	\$1,205	\$1,266	\$1,353
Operations and Maintenance	\$1,066	\$1,030	\$1,058	\$1,035	\$1,055	\$1,105	\$1,158	\$1,160	\$1,161	\$1,185	\$1,301	\$1,302	\$1,482	\$1,566	\$1,783	\$1,845
Other Teaching Services	\$1,025	\$1,073	\$1,133	\$1,141	\$1,232	\$1,284	\$1,342	\$1,413	\$1,449	\$1,508	\$1,553	\$1,582	\$1,714	\$1,895	\$2,100	\$2,190
Pupil Services	\$1,224	\$1,214	\$1,276	\$1,283	\$1,377	\$1,442	\$1,491	\$1,580	\$1,584	\$1,686	\$1,758	\$1,685	\$1,747	\$2,103	\$2,348	\$2,566
Insurance, Retirement and Other	\$2,083	\$2,124	\$2,247	\$2,301	\$2,318	\$2,491	\$2,589	\$2,689	\$2,816	\$2,966	\$3,060	\$3,157	\$3,404	\$3,516	\$3,715	\$3,881
Teachers	\$4,664	\$4,689	\$4,806	\$4,921	\$5,078	\$5,194	\$5,331	\$5,496	\$5,601	\$5,735	\$6,009	\$6,181	\$6,614	\$6,861	\$7,123	\$7,409
Total	\$12,234	\$12,444	\$12,865	\$13,063	\$13,532	\$14,039	\$14,479	\$15,010	\$15,343	\$15,789	\$16,472	\$16,798	\$18,314	\$19,558	\$21,055	\$22,065

FTE Enrollment

Table 17

FTE Series	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
In-District FTE Pupils	1,555	1,534	1,521	1,493	1,469	1,438	1,418	1,372	1,371	1,361	1,352	1,339	1,255	1,266	1,252	1,245
Foundation Enrollment	1,603	1,580	1,563	1,543	1,524	1,505	1,486	1,469	1,442	1,432	1,416	1,408	1,394	1,333	1,322	1,306
Out-of-District FTE Pupils	120	125	129	130	134	138	146	150	152	155	162	161	158	158	162	162

NSS/Ch70 Funding Components (\$)

Table 18

Component	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Ch70 Aid	\$5K	\$5K	\$5K	\$5K	\$6K	\$6K	\$6K	\$6K	\$6K	\$6K	\$6K	\$7K	\$7K	\$8K	\$8K	\$9K
Req NSS (minus Ch70)	\$4K	\$4K	\$4K	\$5K	\$5K	\$5K	\$5K	\$5K	\$5K	\$5K	\$5K	\$6K	\$6K	\$6K	\$6K	\$7K
Actual NSS (minus Req NSS)	\$1K	\$1K	\$1K	\$1K	\$1K	\$1K	\$2K	\$2K	\$2K	\$3K	\$3K	\$3K	\$3K	\$3K	\$3K	\$3K
Total NSS	\$10K	\$11K	\$11K	\$11K	\$12K	\$12K	\$13K	\$13K	\$14K	\$14K	\$15K	\$15K	\$15K	\$17K	\$18K	\$19K

Data: Amherst-Pelham

PPE (\$/pupil), FTE Enrollment, and NSS/Ch70 Funding (\$)

PPE by Category (\$/pupil)

Table 19

Category	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Other	\$646	\$534	\$900	\$417	\$459	\$424	\$599	\$860	\$635	\$384	\$610	\$559	\$477	\$527	\$457	\$799
Guidance, Counseling and Testing	\$525	\$580	\$617	\$684	\$652	\$733	\$778	\$798	\$803	\$891	\$925	\$945	\$1,001	\$1,241	\$1,406	\$1,465
Administration	\$652	\$664	\$687	\$789	\$987	\$907	\$953	\$926	\$996	\$857	\$759	\$780	\$788	\$859	\$928	\$1,277
Instructional Leadership	\$1,289	\$1,337	\$1,209	\$1,397	\$1,390	\$1,545	\$1,429	\$1,437	\$1,541	\$1,555	\$1,544	\$1,501	\$1,436	\$1,472	\$1,644	\$2,336
Operations and Maintenance	\$1,459	\$1,416	\$1,559	\$1,468	\$1,481	\$1,585	\$1,473	\$1,577	\$1,904	\$1,678	\$1,776	\$1,601	\$1,705	\$1,778	\$1,957	\$2,221
Other Teaching Services	\$1,199	\$1,264	\$1,313	\$1,468	\$1,588	\$1,739	\$1,770	\$1,883	\$2,014	\$2,066	\$2,122	\$2,214	\$3,010	\$3,050	\$2,968	\$2,851
Pupil Services	\$1,645	\$1,616	\$1,588	\$1,618	\$1,639	\$1,764	\$1,975	\$2,110	\$2,128	\$2,218	\$2,236	\$1,957	\$2,950	\$2,539	\$2,771	\$2,766
Insurance, Retirement and Other	\$3,305	\$3,636	\$3,673	\$3,811	\$3,526	\$3,954	\$4,086	\$4,103	\$4,218	\$5,006	\$5,047	\$5,074	\$5,237	\$5,062	\$5,887	\$6,248
Teachers	\$5,491	\$5,511	\$6,254	\$6,199	\$6,533	\$6,765	\$6,704	\$6,590	\$6,927	\$7,208	\$7,285	\$7,155	\$7,592	\$7,404	\$8,158	\$8,270
Total	\$16,211	\$16,558	\$17,800	\$17,851	\$18,255	\$19,416	\$19,767	\$20,284	\$21,166	\$21,863	\$22,304	\$21,786	\$24,196	\$23,932	\$26,176	\$28,233

FTE Enrollment

Table 20

FTE Series	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
In-District FTE Pupils	1,712	1,659	1,573	1,536	1,519	1,470	1,450	1,378	1,366	1,342	1,334	1,346	1,278	1,265	1,241	1,209
Foundation Enrollment	1,801	1,763	1,726	1,647	1,610	1,581	1,529	1,532	1,496	1,490	1,433	1,399	1,399	1,330	1,310	1,274
Out-of-District FTE Pupils	112	129	138	143	138	149	169	190	192	170	158	152	155	145	150	147

NSS/Ch70 Funding Components (\$)

Table 21

Component	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Ch70 Aid	\$5K	\$5K	\$5K	\$6K	\$6K	\$6K	\$6K	\$6K	\$6K	\$6K	\$7K	\$7K	\$7K	\$7K	\$7K	\$8K
Req NSS (minus Ch70)	\$4K	\$6K	\$6K	\$7K	\$7K	\$7K	\$8K	\$8K	\$8K	\$8K	\$8K	\$9K	\$9K	\$10K	\$10K	\$11K
Actual NSS (minus Req NSS)	\$3K	\$2K	\$3K	\$3K	\$4K	\$4K	\$4K	\$4K	\$5K	\$5K	\$5K	\$5K	\$5K	\$5K	\$5K	\$6K
Total NSS	\$13K	\$14K	\$14K	\$16K	\$17K	\$17K	\$18K	\$18K	\$19K	\$19K	\$20K	\$21K	\$21K	\$22K	\$23K	\$25K

Data: Amherst

PPE (\$/pupil), FTE Enrollment, and NSS/Ch70 Funding (\$)

PPE by Category (\$/pupil)

Table 22

Category	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Other	\$684	\$547	\$606	\$931	\$1,098	\$1,098	\$1,265	\$1,242	\$1,119	\$798	\$1,092	\$733	\$1,563	\$1,561	\$1,475	\$1,328
Guidance, Counseling and Testing	\$396	\$383	\$362	\$411	\$469	\$538	\$545	\$583	\$599	\$631	\$668	\$696	\$830	\$891	\$1,093	\$1,053
Administration	\$627	\$698	\$735	\$709	\$787	\$809	\$919	\$944	\$981	\$799	\$767	\$827	\$874	\$896	\$981	\$1,213
Instructional Leadership	\$1,134	\$1,079	\$965	\$1,032	\$1,153	\$1,275	\$1,175	\$1,131	\$1,087	\$1,231	\$1,325	\$1,527	\$1,750	\$1,783	\$1,847	\$2,498
Operations and Maintenance	\$1,264	\$1,271	\$1,636	\$1,436	\$1,399	\$1,451	\$1,577	\$1,530	\$1,555	\$1,340	\$1,646	\$1,764	\$1,897	\$1,874	\$1,941	\$2,248
Other Teaching Services	\$1,762	\$1,813	\$1,923	\$2,345	\$2,621	\$2,610	\$2,840	\$2,652	\$2,751	\$2,973	\$3,084	\$3,053	\$3,908	\$3,898	\$3,746	\$4,612
Pupil Services	\$865	\$1,003	\$973	\$975	\$1,084	\$1,223	\$1,257	\$1,267	\$1,219	\$1,278	\$1,362	\$1,148	\$1,205	\$1,849	\$1,728	\$1,965
Insurance, Retirement and Other	\$3,353	\$3,610	\$3,821	\$3,852	\$3,941	\$4,189	\$4,201	\$4,295	\$4,231	\$4,840	\$5,212	\$5,363	\$5,661	\$5,561	\$6,022	\$6,605
Teachers	\$5,944	\$6,213	\$6,355	\$6,963	\$6,804	\$6,627	\$6,841	\$7,712	\$7,792	\$8,091	\$8,538	\$8,725	\$9,528	\$9,598	\$9,451	\$9,745
Total	\$16,029	\$16,617	\$17,376	\$18,654	\$19,356	\$19,820	\$20,620	\$21,356	\$21,334	\$21,981	\$23,694	\$23,836	\$27,216	\$27,911	\$28,284	\$31,267

FTE Enrollment

Table 23

FTE Series	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
In-District FTE Pupils	1,387	1,331	1,284	1,215	1,196	1,197	1,184	1,154	1,156	1,163	1,131	1,107	1,017	1,052	1,062	1,002
Foundation Enrollment	1,398	1,405	1,356	1,279	1,269	1,227	1,205	1,192	1,199	1,136	1,106	1,134	1,097	1,045	1,065	1,055
Out-of-District FTE Pupils	65	72	73	84	90	90	101	113	110	108	124	122	118	123	129	134

NSS/Ch70 Funding Components (\$)

Table 24

Component	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Ch70 Aid	\$4K	\$4K	\$4K	\$5K	\$5K	\$5K	\$5K	\$5K	\$5K	\$5K	\$6K	\$5K	\$6K	\$6K	\$6K	\$6K
Req NSS (minus Ch70)	\$4K	\$6K	\$6K	\$7K	\$7K	\$7K	\$7K	\$8K	\$8K	\$8K	\$8K	\$8K	\$9K	\$9K	\$10K	\$11K
Actual NSS (minus Req NSS)	\$6K	\$5K	\$5K	\$6K	\$6K	\$6K	\$7K	\$8K	\$8K	\$9K	\$10K	\$9K	\$9K	\$11K	\$11K	\$11K
Total NSS	\$15K	\$15K	\$16K	\$17K	\$17K	\$18K	\$19K	\$20K	\$20K	\$22K	\$23K	\$23K	\$24K	\$26K	\$26K	\$28K

Data: Leverett

PPE (\$/pupil), FTE Enrollment, and NSS/Ch70 Funding (\$)

PPE by Category (\$/pupil)

Table 25

Category	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Other	\$575	\$602	\$651	\$766	\$616	\$556	\$563	\$529	\$596	\$453	\$503	\$606	\$1,041	\$840	\$756	\$608
Guidance, Counseling and Testing	\$310	\$306	\$314	\$350	\$354	\$412	\$292	\$304	\$323	\$355	\$351	\$362	\$498	\$790	\$659	\$666
Administration	\$715	\$676	\$710	\$888	\$925	\$1,029	\$809	\$842	\$1,244	\$899	\$826	\$857	\$1,010	\$987	\$1,061	\$953
Instructional Leadership	\$1,015	\$1,008	\$1,072	\$1,244	\$1,257	\$1,393	\$1,357	\$1,344	\$1,431	\$1,461	\$1,278	\$1,317	\$1,318	\$1,354	\$1,430	\$1,663
Operations and Maintenance	\$1,755	\$1,427	\$1,532	\$1,741	\$1,479	\$2,421	\$2,272	\$1,707	\$1,209	\$1,765	\$1,491	\$1,458	\$2,131	\$2,296	\$2,292	\$2,158
Other Teaching Services	\$2,385	\$2,040	\$2,074	\$2,346	\$2,540	\$2,352	\$2,274	\$2,281	\$2,193	\$2,175	\$2,477	\$2,386	\$2,178	\$3,119	\$3,266	\$3,526
Pupil Services	\$1,085	\$664	\$782	\$1,249	\$1,259	\$1,980	\$1,781	\$1,397	\$1,759	\$1,726	\$1,494	\$1,279	\$2,192	\$1,120	\$1,976	\$2,148
Insurance, Retirement and Other	\$2,098	\$2,041	\$2,253	\$2,499	\$2,753	\$3,245	\$3,041	\$2,937	\$2,943	\$3,677	\$3,781	\$3,686	\$4,766	\$3,630	\$3,611	\$3,793
Teachers	\$5,218	\$5,690	\$5,818	\$6,301	\$6,138	\$7,547	\$6,868	\$6,967	\$6,814	\$7,967	\$7,665	\$7,134	\$9,224	\$6,888	\$6,724	\$6,395
Total	\$15,156	\$14,454	\$15,206	\$17,384	\$17,321	\$20,935	\$19,257	\$18,308	\$18,512	\$20,478	\$19,866	\$19,085	\$24,358	\$21,024	\$21,775	\$21,910

FTE Enrollment

Table 26

FTE Series	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
In-District FTE Pupils	162	164	161	150	147	124	134	135	134	129	136	142	117	142	140	142
Foundation Enrollment	125	120	123	123	119	118	100	104	101	123	103	112	116	105	112	101
Out-of-District FTE Pupils	7	7	6	9	10	9	8	6	14	8	9	12	9	12	7	3

NSS/Ch70 Funding Components (\$)

Table 27

Component	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Ch70 Aid	\$2K	\$2K	\$2K	\$2K	\$2K	\$2K	\$3K	\$3K	\$3K	\$2K	\$3K	\$3K	\$3K	\$3K	\$3K	\$4K
Req NSS (minus Ch70)	\$7K	\$7K	\$8K	\$8K	\$8K	\$8K	\$8K	\$8K	\$8K	\$8K	\$8K	\$8K	\$9K	\$9K	\$9K	\$9K
Actual NSS (minus Req NSS)	\$7K	\$7K	\$7K	\$8K	\$9K	\$8K	\$10K	\$10K	\$10K	\$8K	\$11K	\$11K	\$10K	\$13K	\$10K	\$11K
Total NSS	\$16K	\$17K	\$17K	\$17K	\$19K	\$18K	\$21K	\$21K	\$21K	\$18K	\$23K	\$22K	\$21K	\$25K	\$22K	\$24K

Data: Pelham

PPE (\$/pupil), FTE Enrollment, and NSS/Ch70 Funding (\$)

PPE by Category (\$/pupil)

Table 28

Category	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Other	\$580	\$361	\$384	\$366	\$658	\$639	\$780	\$652	\$569	\$414	\$466	\$407	\$292	\$365	\$422	\$207
Guidance, Counseling and Testing	\$438	\$449	\$459	\$449	\$617	\$664	\$696	\$683	\$682	\$499	\$499	\$582	\$795	\$772	\$775	\$1,855
Administration	\$390	\$398	\$478	\$391	\$664	\$796	\$806	\$980	\$739	\$731	\$489	\$574	\$656	\$688	\$614	\$917
Instructional Leadership	\$1,123	\$1,159	\$1,089	\$1,064	\$1,113	\$1,252	\$1,288	\$1,322	\$1,278	\$1,431	\$1,194	\$1,609	\$1,670	\$1,873	\$1,782	\$1,885
Operations and Maintenance	\$920	\$847	\$925	\$884	\$992	\$947	\$1,080	\$942	\$987	\$889	\$1,229	\$1,304	\$1,498	\$1,380	\$1,524	\$1,614
Other Teaching Services	\$2,184	\$2,109	\$2,166	\$2,026	\$1,937	\$2,063	\$2,218	\$2,541	\$2,850	\$2,269	\$2,273	\$2,324	\$2,540	\$2,292	\$2,594	\$3,010
Pupil Services	\$594	\$641	\$888	\$1,058	\$967	\$951	\$1,000	\$1,114	\$1,092	\$1,173	\$1,233	\$1,060	\$1,317	\$1,431	\$1,113	\$1,755
Insurance, Retirement and Other	\$2,615	\$2,585	\$2,602	\$2,897	\$3,052	\$3,283	\$3,376	\$3,473	\$3,863	\$4,561	\$3,791	\$4,168	\$4,850	\$4,157	\$3,983	\$4,670
Teachers	\$5,889	\$5,474	\$5,627	\$5,576	\$5,035	\$6,090	\$6,540	\$6,418	\$5,904	\$6,310	\$6,129	\$6,860	\$7,406	\$6,714	\$6,430	\$7,127
Total	\$14,733	\$14,023	\$14,618	\$14,711	\$15,035	\$16,685	\$17,784	\$18,125	\$17,964	\$18,277	\$17,303	\$18,888	\$21,024	\$19,672	\$19,237	\$23,040

FTE Enrollment

Table 29

FTE Series	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
In-District FTE Pupils	118	125	122	128	132	128	123	123	130	127	134	125	106	119	134	126
Foundation Enrollment	85	81	81	68	63	67	70	73	74	81	83	82	84	70	65	63
Out-of-District FTE Pupils	3	3	4	1	4	7	5	7	11	12	10	4	8	6	5	9

NSS/Ch70 Funding Components (\$)

Table 30

Component	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Ch70 Aid	\$3K	\$3K	\$3K	\$3K	\$3K	\$3K	\$3K	\$3K	\$3K	\$3K	\$3K	\$3K	\$3K	\$3K	\$4K	\$4K
Req NSS (minus Ch70)	\$5K	\$6K	\$6K	\$7K	\$8K	\$8K	\$9K	\$9K	\$8K	\$8K	\$8K	\$8K	\$8K	\$9K	\$10K	\$10K
Actual NSS (minus Req NSS)	\$8K	\$8K	\$8K	\$10K	\$12K	\$10K	\$9K	\$10K	\$12K	\$12K	\$11K	\$11K	\$13K	\$16K	\$19K	\$20K
Total NSS	\$16K	\$17K	\$17K	\$20K	\$23K	\$22K	\$21K	\$22K	\$23K	\$23K	\$22K	\$23K	\$24K	\$29K	\$32K	\$34K

Data: Shutesbury

PPE (\$/pupil), FTE Enrollment, and NSS/Ch70 Funding (\$)

PPE by Category (\$/pupil)

Table 31

Category	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Other	\$580	\$498	\$564	\$865	\$704	\$515	\$636	\$684	\$1,510	\$420	\$385	\$873	\$1,296	\$551	\$1,102	\$545
Guidance, Counseling and Testing	\$302	\$328	\$345	\$334	\$345	\$344	\$354	\$434	\$318	\$546	\$542	\$383	\$519	\$1,393	\$697	\$826
Administration	\$652	\$943	\$767	\$836	\$877	\$726	\$884	\$1,084	\$1,040	\$846	\$856	\$938	\$925	\$1,100	\$934	\$1,015
Instructional Leadership	\$969	\$1,138	\$1,149	\$1,198	\$1,227	\$1,003	\$1,192	\$1,517	\$1,604	\$1,481	\$1,517	\$1,541	\$1,707	\$1,736	\$1,323	\$2,103
Operations and Maintenance	\$1,180	\$1,079	\$1,221	\$1,276	\$1,251	\$1,338	\$1,384	\$1,637	\$1,791	\$1,941	\$1,663	\$1,916	\$2,665	\$2,048	\$2,051	\$2,754
Other Teaching Services	\$2,185	\$2,447	\$2,361	\$2,193	\$2,179	\$2,591	\$2,164	\$2,817	\$3,301	\$3,491	\$3,545	\$3,560	\$3,722	\$5,320	\$3,441	\$4,359
Pupil Services	\$990	\$800	\$850	\$1,222	\$1,283	\$1,074	\$1,073	\$906	\$1,847	\$1,755	\$1,664	\$1,815	\$1,937	\$1,355	\$2,249	\$2,136
Insurance, Retirement and Other	\$2,332	\$2,687	\$2,596	\$2,690	\$2,564	\$2,484	\$2,328	\$3,107	\$3,773	\$4,388	\$2,970	\$3,963	\$3,350	\$3,154	\$4,007	\$3,501
Teachers	\$4,821	\$5,363	\$5,603	\$5,169	\$5,509	\$6,012	\$5,620	\$6,612	\$7,043	\$7,838	\$7,914	\$8,204	\$8,933	\$10,075	\$8,898	\$9,669
Total	\$14,011	\$15,283	\$15,456	\$15,783	\$15,939	\$16,087	\$15,635	\$18,798	\$22,227	\$22,706	\$21,056	\$23,193	\$25,054	\$26,732	\$24,702	\$26,908

FTE Enrollment

Table 32

FTE Series	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
In-District FTE Pupils	162	151	146	151	150	157	156	131	120	124	127	125	113	115	123	110
Foundation Enrollment	156	149	141	138	143	132	141	146	118	116	118	117	105	98	100	111
Out-of-District FTE Pupils	7	5	7	10	9	13	9	12	13	16	18	15	16	18	19	18

NSS/Ch70 Funding Components (\$)

Table 33

Component	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Ch70 Aid	\$4K	\$4K	\$4K	\$4K	\$4K	\$4K	\$4K	\$4K	\$5K	\$5K	\$5K	\$5K	\$6K	\$6K	\$6K	\$6K
Req NSS (minus Ch70)	\$4K	\$5K	\$5K	\$5K	\$5K	\$5K	\$5K	\$5K	\$5K	\$6K	\$6K	\$6K	\$7K	\$7K	\$7K	\$7K
Actual NSS (minus Req NSS)	\$4K	\$5K	\$6K	\$6K	\$6K	\$7K	\$7K	\$6K	\$9K	\$10K	\$10K	\$12K	\$12K	\$15K	\$13K	\$10K
Total NSS	\$13K	\$14K	\$15K	\$15K	\$15K	\$17K	\$16K	\$15K	\$20K	\$22K	\$21K	\$23K	\$25K	\$29K	\$26K	\$24K

Appendix D. Additional Visualizations

Scatterplot of enrollment vs. per-pupil expenditure with quartile boundaries (2019)

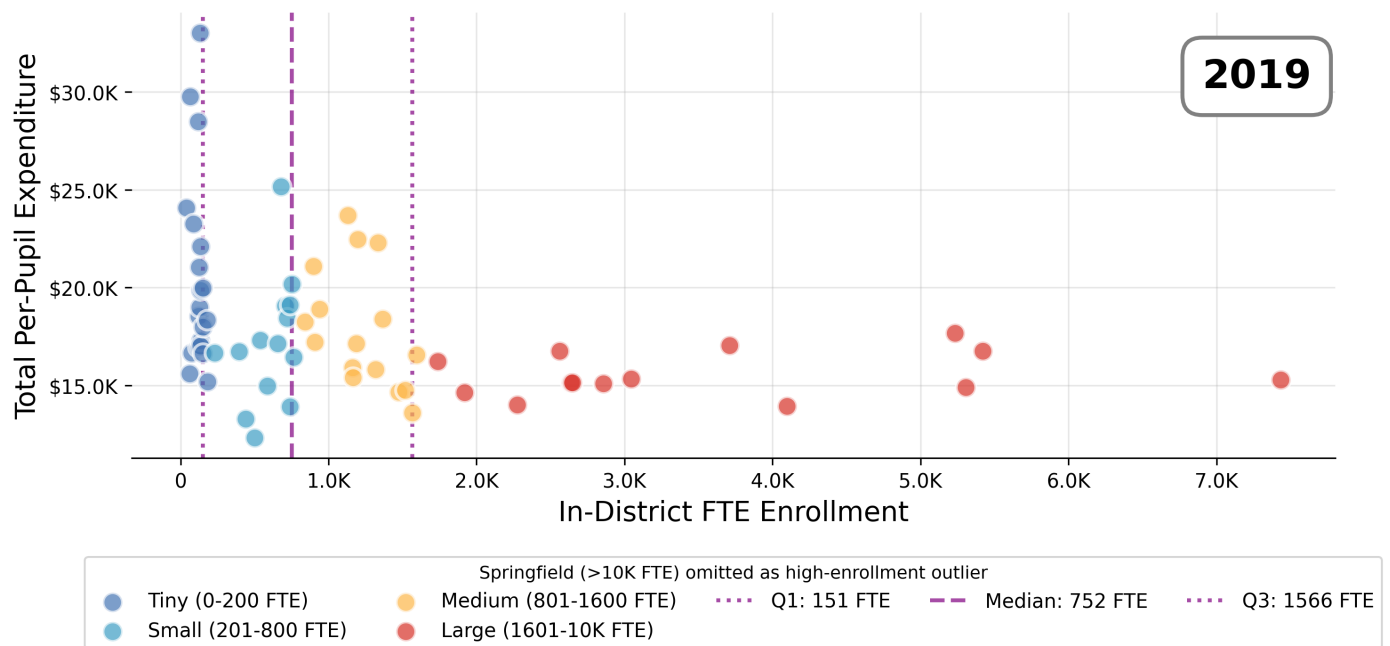


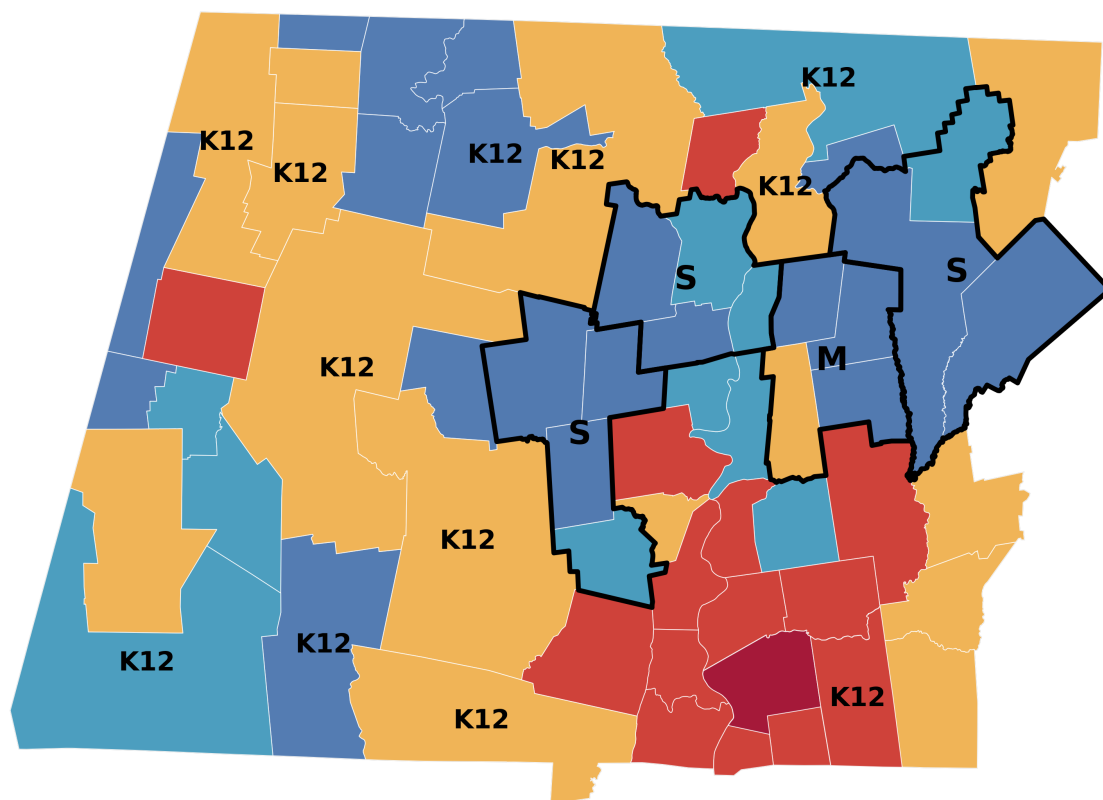
Figure 1

This appendix contains historical scatterplots and geographic maps showing the evolution of district enrollment and per-pupil expenditures over time. These visualizations provide context for understanding how enrollment patterns and spending levels have shifted across Western Massachusetts districts from 2009 to 2024.

This scatterplot shows the relationship between district enrollment (x-axis) and per-pupil expenditure (y-axis) for Western Massachusetts traditional districts in 2019. Each point represents one district, colored by enrollment cohort. Horizontal lines show quartile boundaries used to define cohorts.

Appendix D. Additional Visualizations (continued)

Geographic map showing district locations and enrollment cohorts (2019)



Enrollment Cohorts - 2019

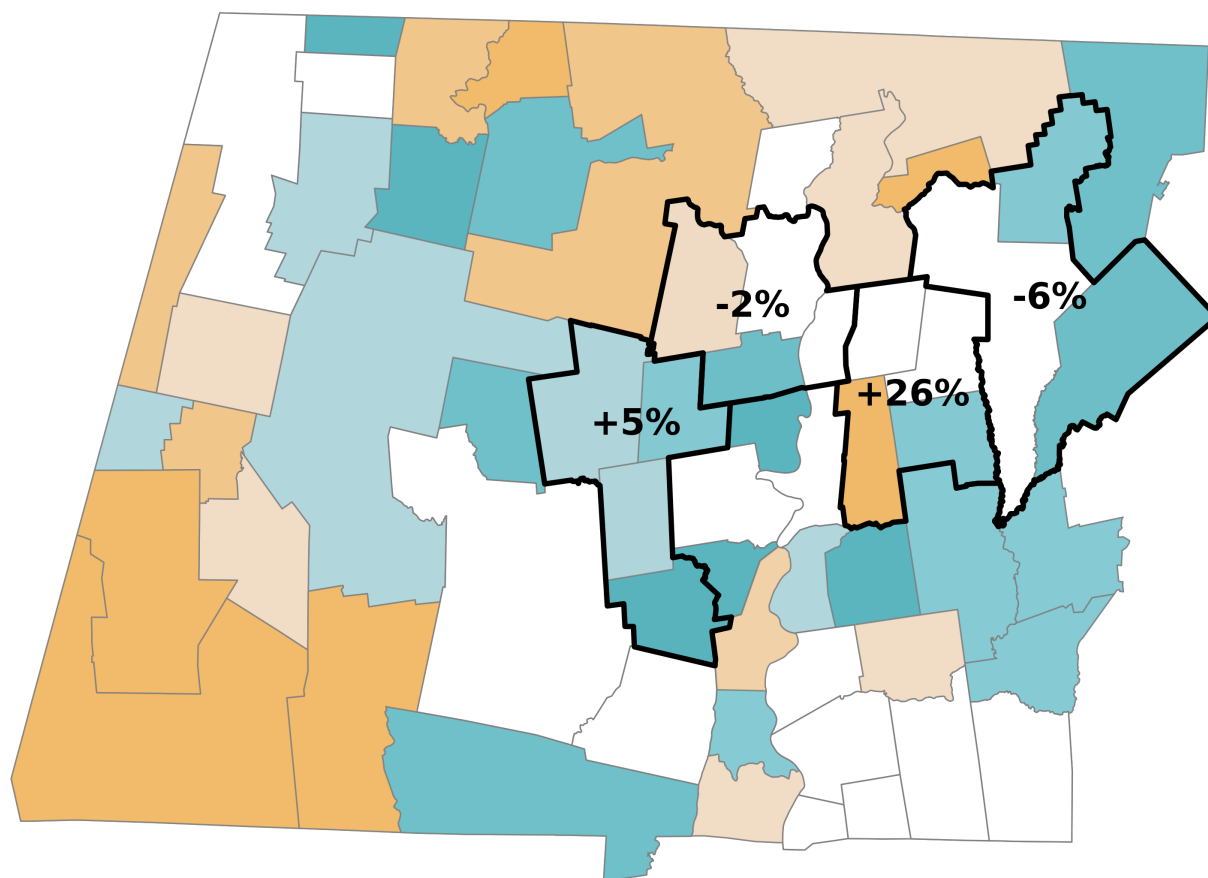
■ Tiny (0-200 FTE): 20 district(s)	 Secondary regional (black border)
■ Small (201-800 FTE): 14 district(s)	Cohort: T, S, M, L, XL
■ Medium (801-1600 FTE): 16 district(s)	K12 = Regional K-12 with all grade levels in a single district
■ Large (1601-10K FTE): 14 district(s)	
■ Outliers (Springfield >10K FTE): 1 district(s)	

Figure 2

Geography matters for understanding school costs. This map shows where enrollment cohorts cluster and how regional district boundaries cross municipal lines.

Appendix D. Additional Visualizations (continued)

Geographic map showing 2019 PPE vs enrollment cohort baseline



Total PPE vs Cohort Baseline - 2019 ($\pm 5\%$ threshold)

■ $\leq -20\%$ cohort avg: 6 district(s)	■ +5 to +10% cohort avg: 8 district(s)
■ -15 to -20% cohort avg: 6 district(s)	■ +10 to +15% cohort avg: 1 district(s)
■ -10 to -15% cohort avg: 7 district(s)	■ +15 to +20% cohort avg: 4 district(s)
■ -5 to -10% cohort avg: 7 district(s)	■ $\geq +20\%$ cohort avg: 7 district(s)
■ Within $\pm 5\%$ cohort avg: 19 district(s)	+/-% = Secondary regional district deviation (n=4)

Figure 3

This map shows each district's 2024 per-pupil spending compared to its enrollment cohort baseline (weighted average of all Western MA districts in that cohort). This helps identify which districts are operating above or below the typical spending level for their enrollment size, accounting for the well-documented relationship between district size and per-pupil costs.

Appendix D. Additional Visualizations (continued)

Geographic map showing 10-year PPE growth (2009-2019) vs enrollment cohort baseline

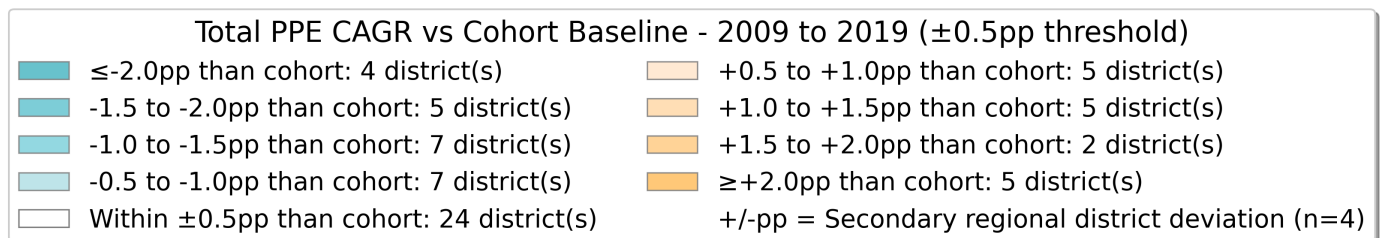
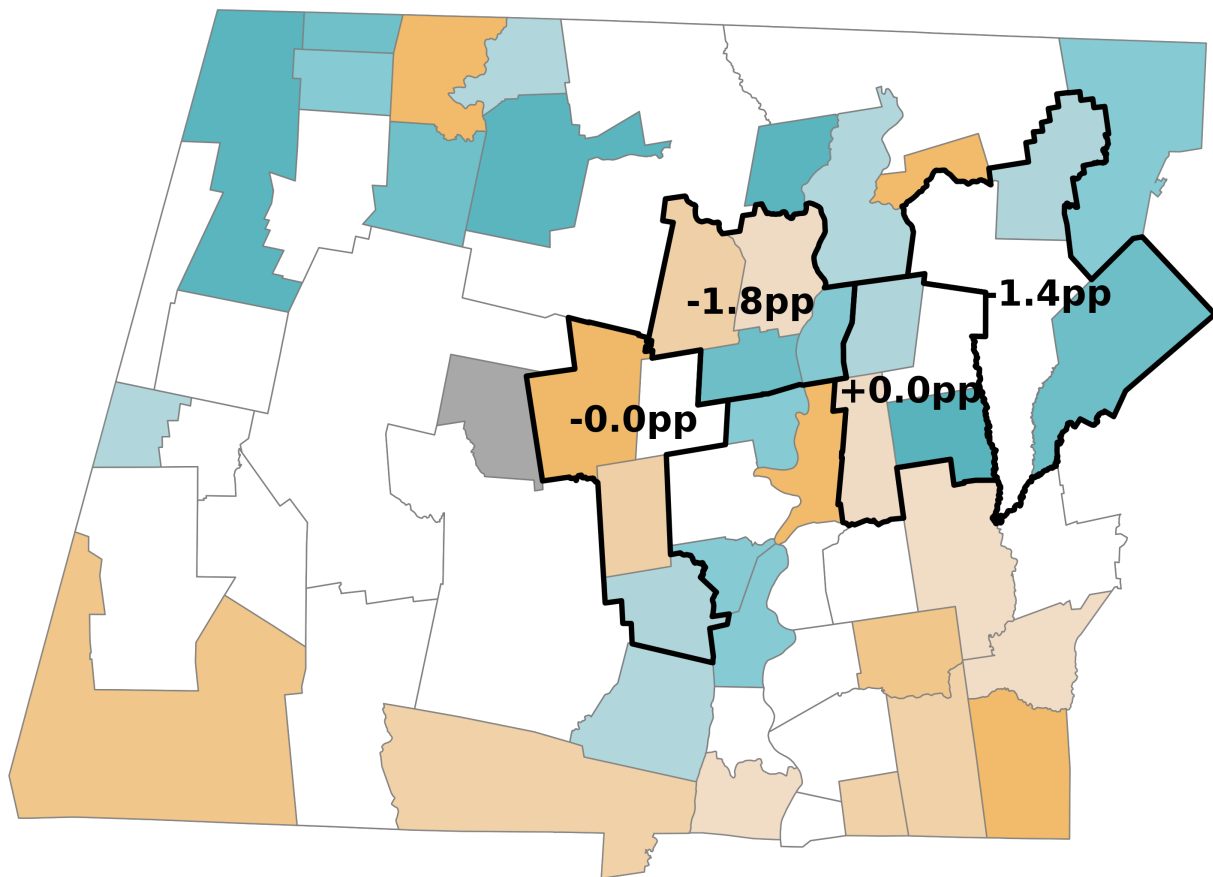


Figure 4

This map shows each district's 15-year spending growth rate (CAGR, 2009-2024) compared to its enrollment cohort baseline growth rate. This helps identify which districts experienced faster or slower spending growth than similar-sized districts, revealing local policy decisions and cost pressures beyond typical enrollment cohort trends.

Appendix D. Additional Visualizations (continued)

Scatterplot of enrollment vs. per-pupil expenditure with quartile boundaries (2014)

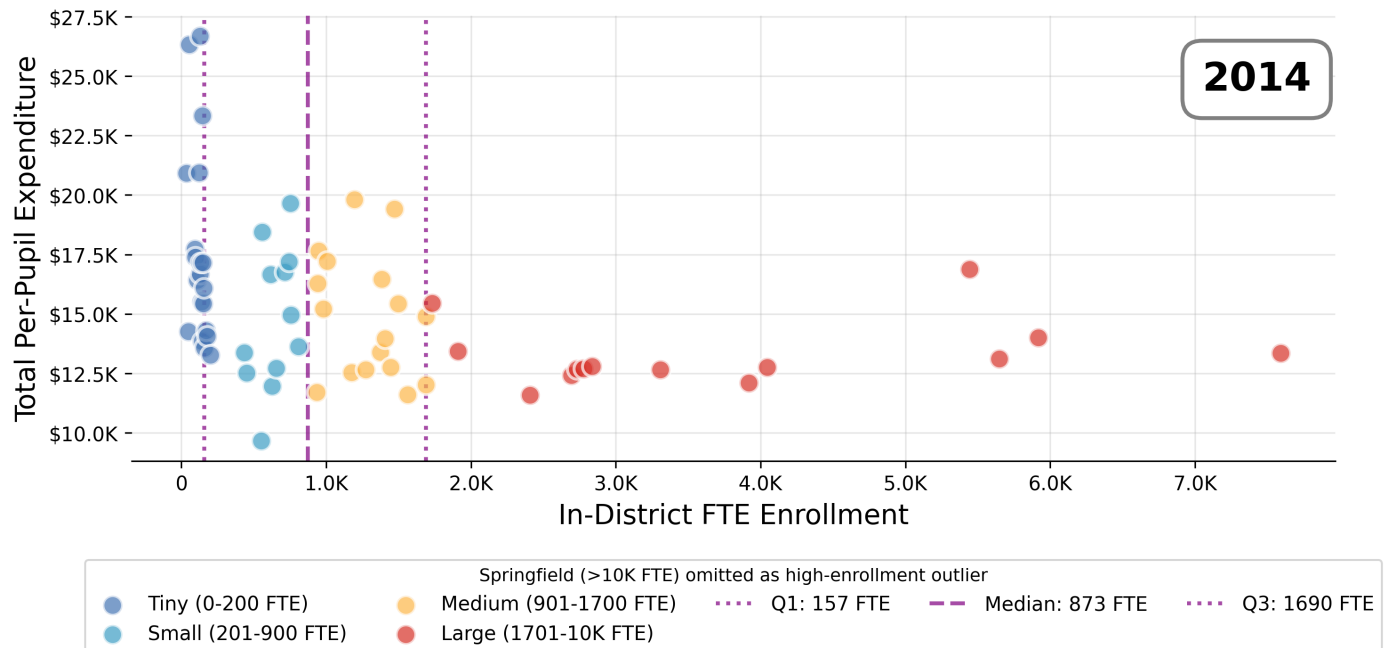


Figure 5

This scatterplot shows the relationship between district enrollment (x-axis) and per-pupil expenditure (y-axis) for Western Massachusetts traditional districts in 2014. Each point represents one district, colored by enrollment cohort. Horizontal lines show quartile boundaries used to define cohorts.

Appendix D. Additional Visualizations (continued)

Geographic map showing district locations and enrollment cohorts (2014)

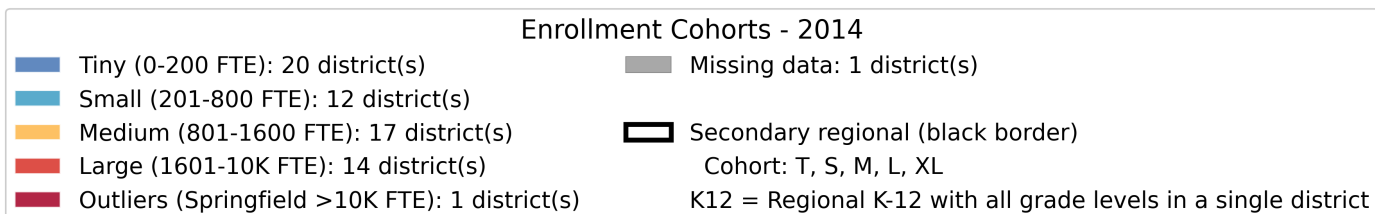
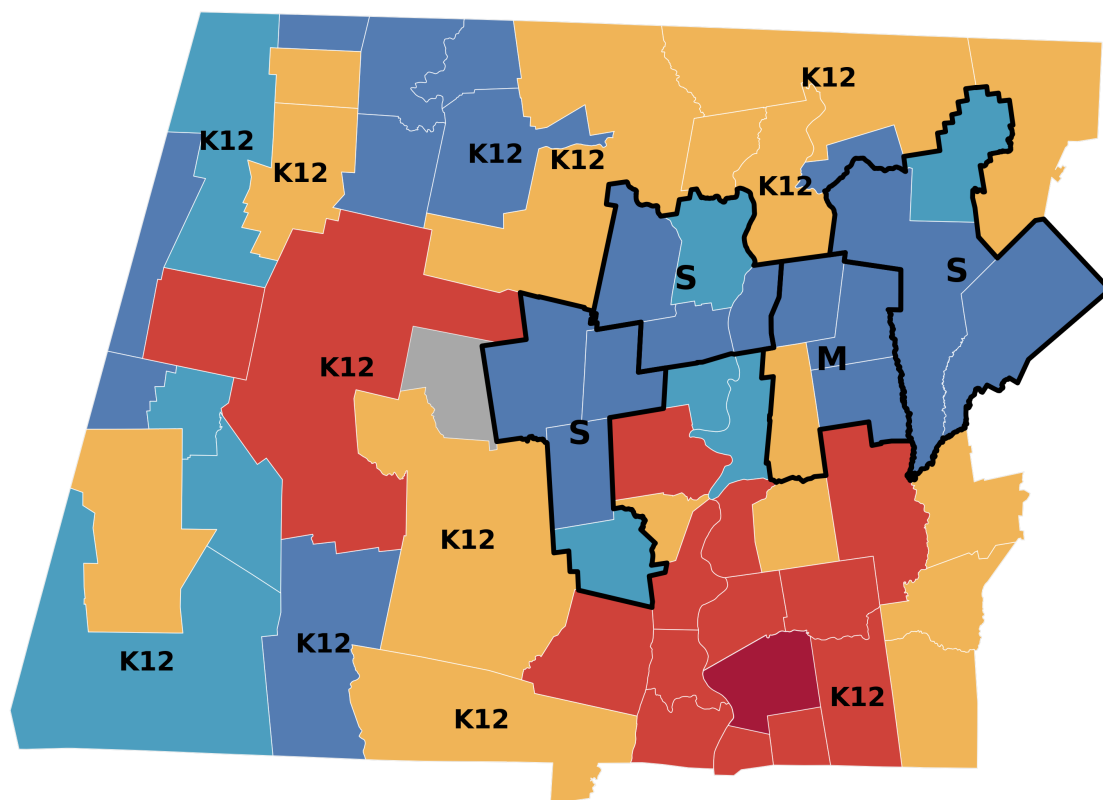


Figure 6

Geography matters for understanding school costs. This map shows where enrollment cohorts cluster and how regional district boundaries cross municipal lines.

Appendix D. Additional Visualizations (continued)

Geographic map showing 2014 PPE vs enrollment cohort baseline

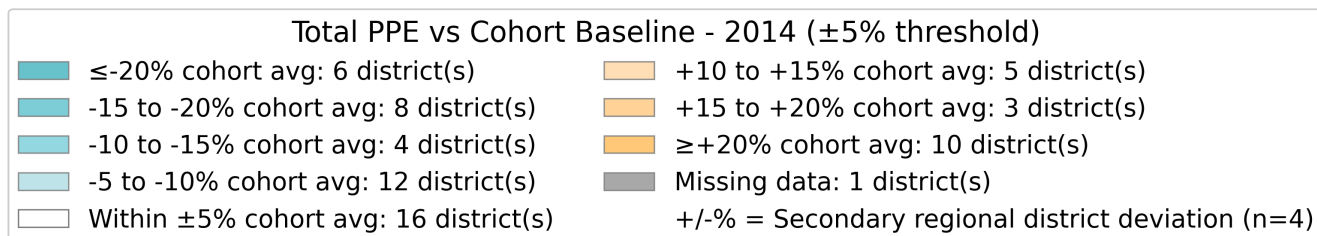
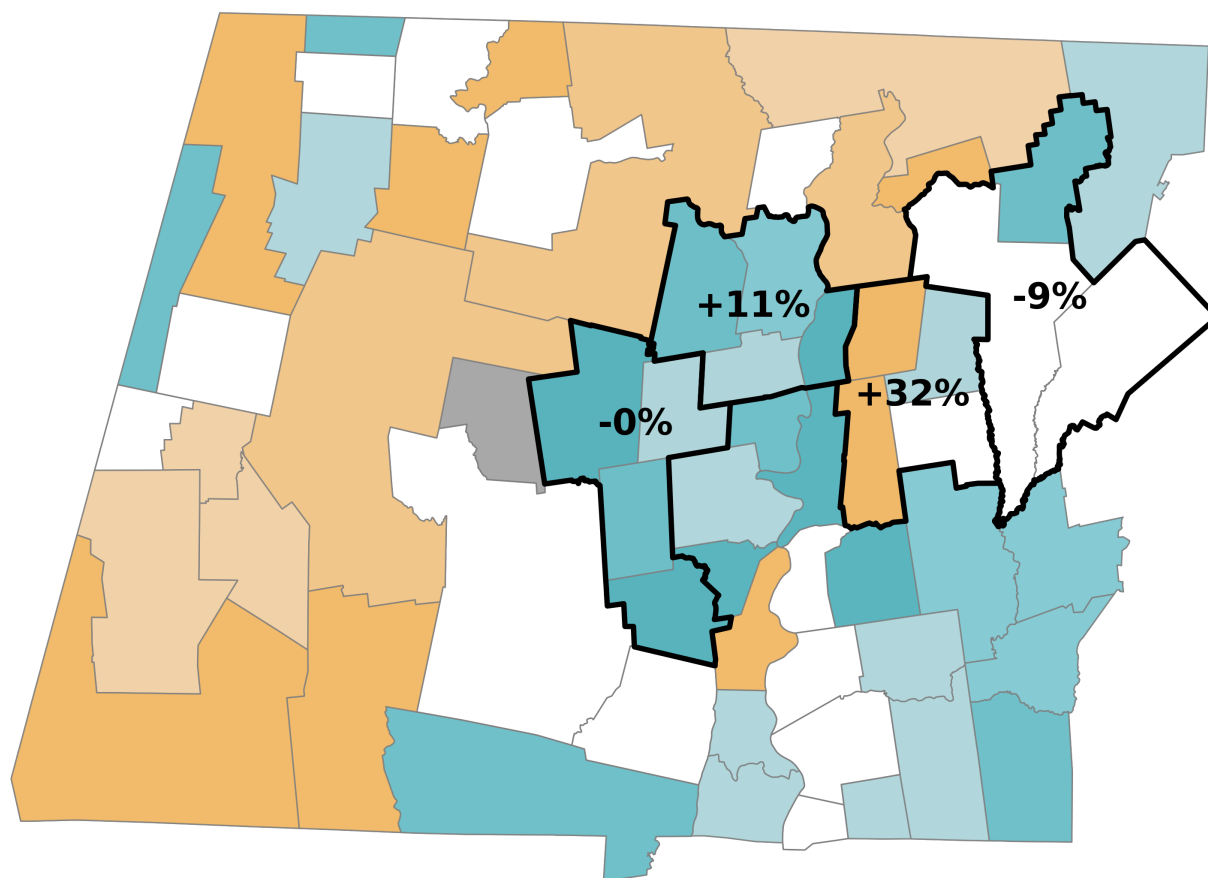


Figure 7

This map shows each district's 2024 per-pupil spending compared to its enrollment cohort baseline (weighted average of all Western MA districts in that cohort). This helps identify which districts are operating above or below the typical spending level for their enrollment size, accounting for the well-documented relationship between district size and per-pupil costs.

Appendix D. Additional Visualizations (continued)

Geographic map showing 5-year PPE growth (2009-2014) vs enrollment cohort baseline

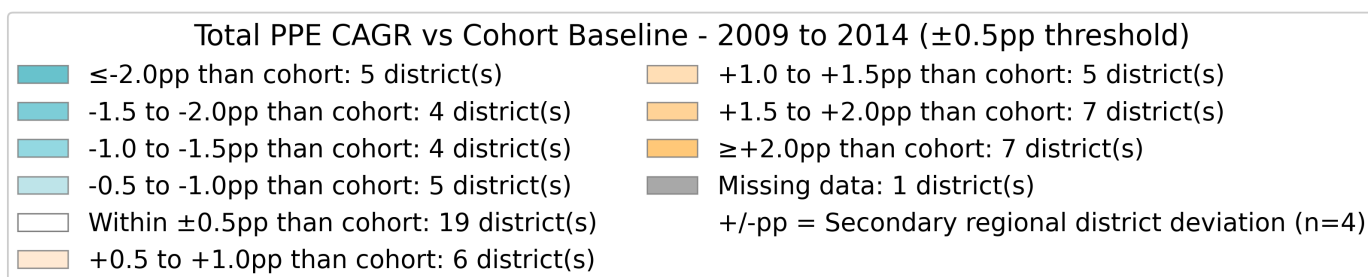
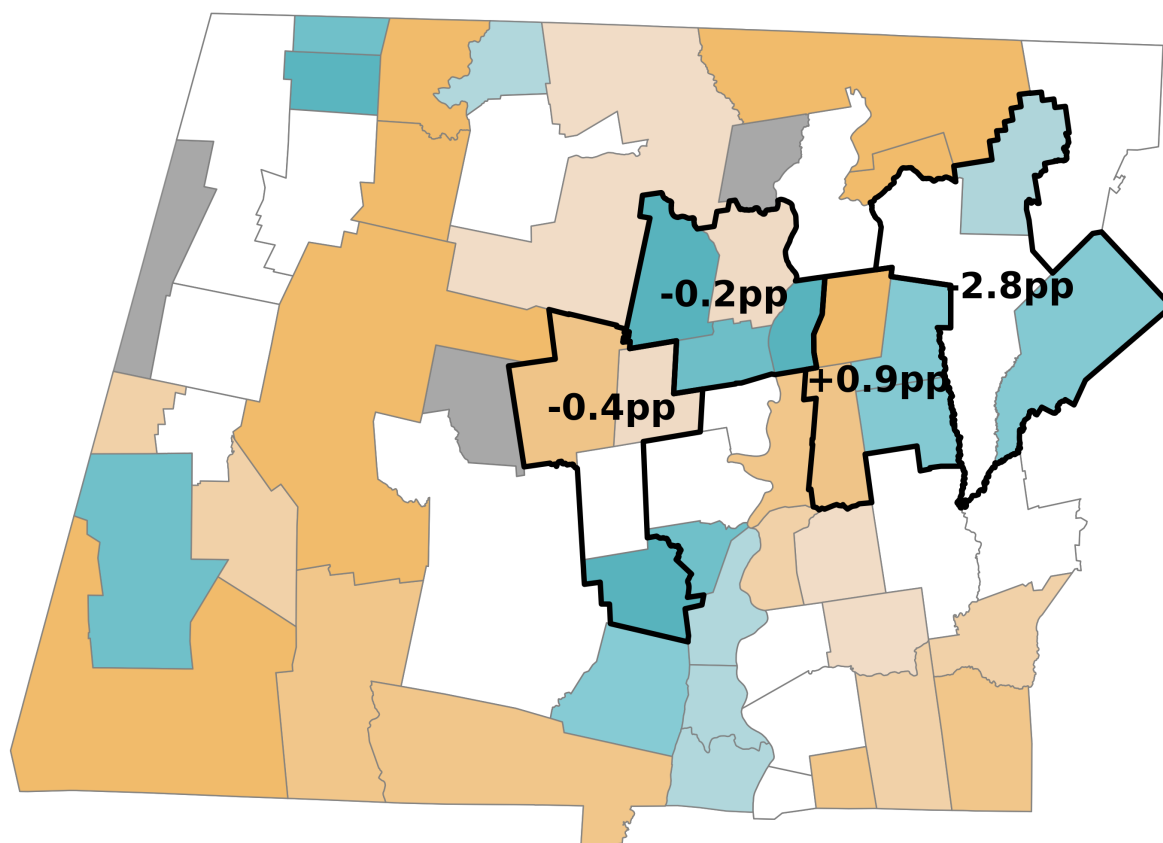


Figure 8

This map shows each district's 15-year spending growth rate (CAGR, 2009-2024) compared to its enrollment cohort baseline growth rate. This helps identify which districts experienced faster or slower spending growth than similar-sized districts, revealing local policy decisions and cost pressures beyond typical enrollment cohort trends.

Appendix D. Additional Visualizations (continued)

Scatterplot of enrollment vs. per-pupil expenditure with quartile boundaries (2009)

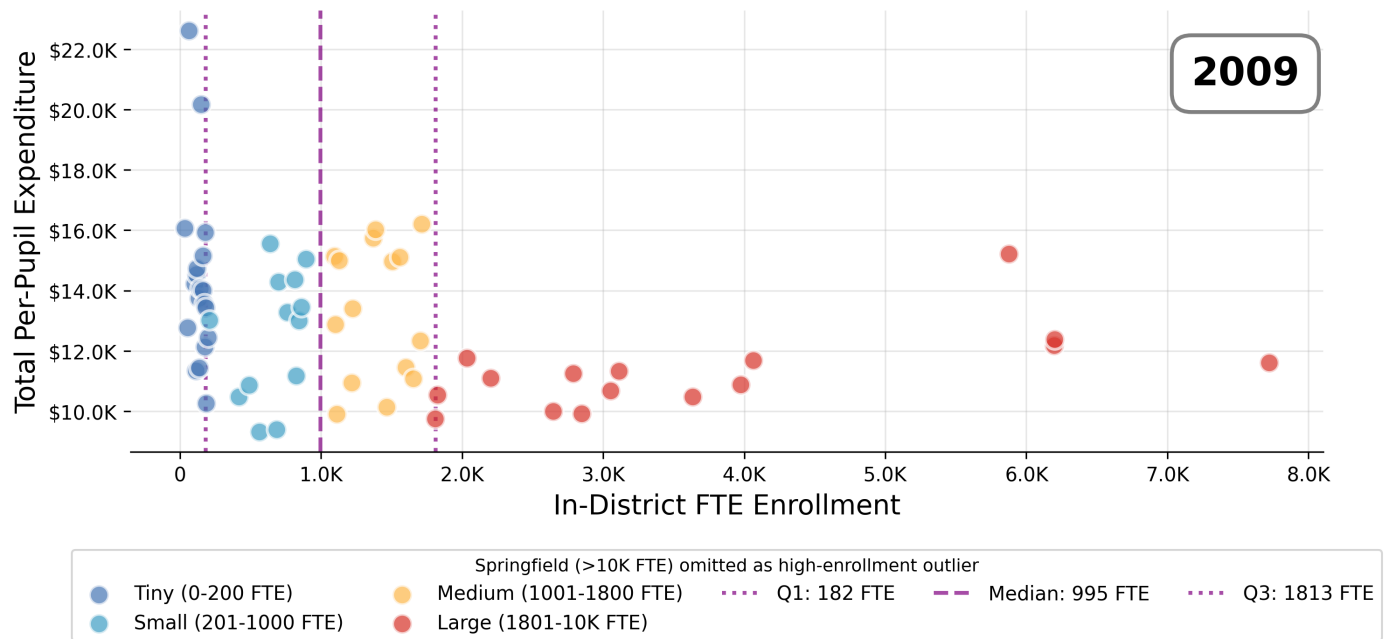


Figure 9

This scatterplot shows the relationship between district enrollment (x-axis) and per-pupil expenditure (y-axis) for Western Massachusetts traditional districts in 2009. Each point represents one district, colored by enrollment cohort. Horizontal lines show quartile boundaries used to define cohorts.

Appendix D. Additional Visualizations (continued)

Geographic map showing district locations and enrollment cohorts (2009)

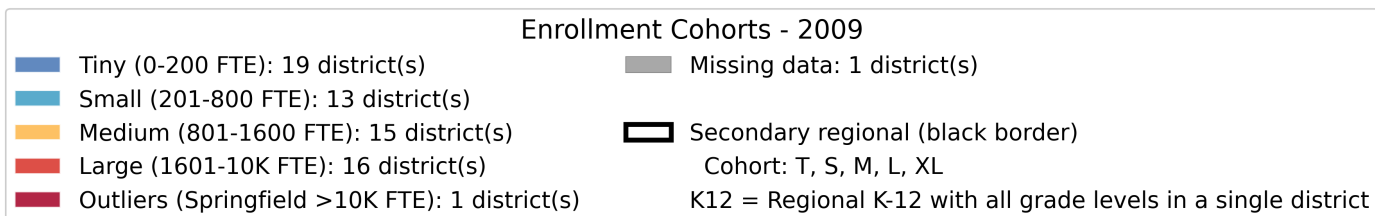
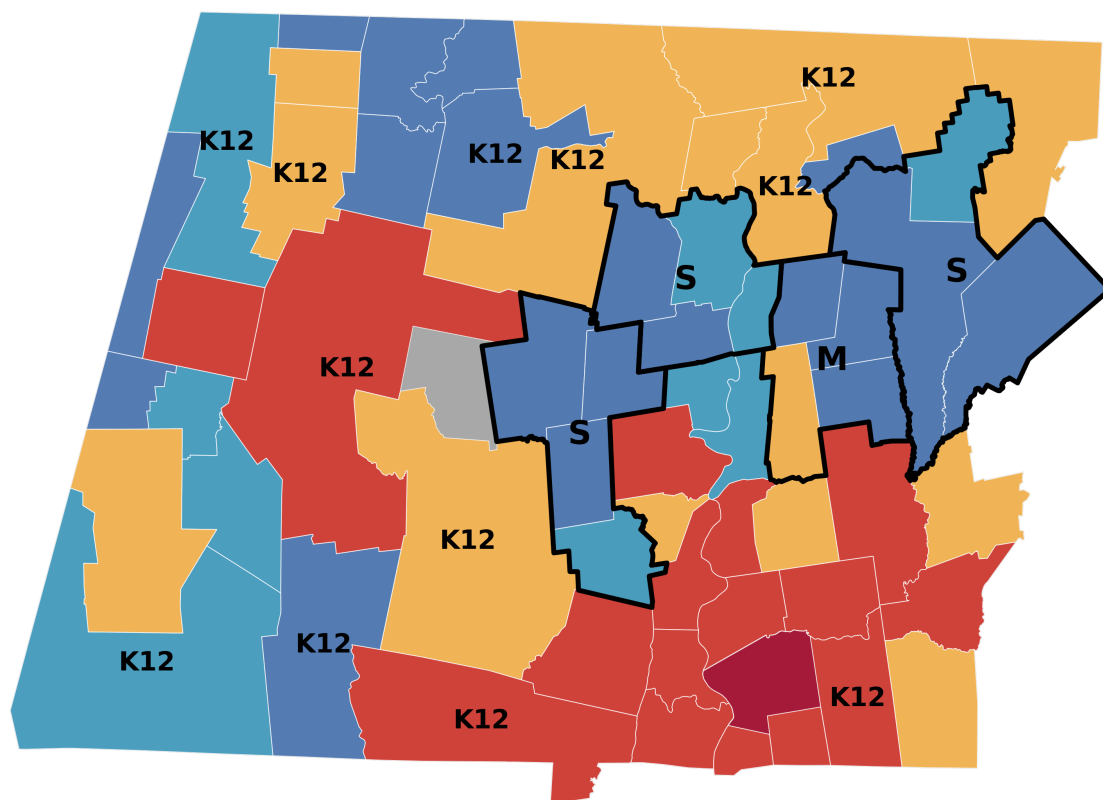
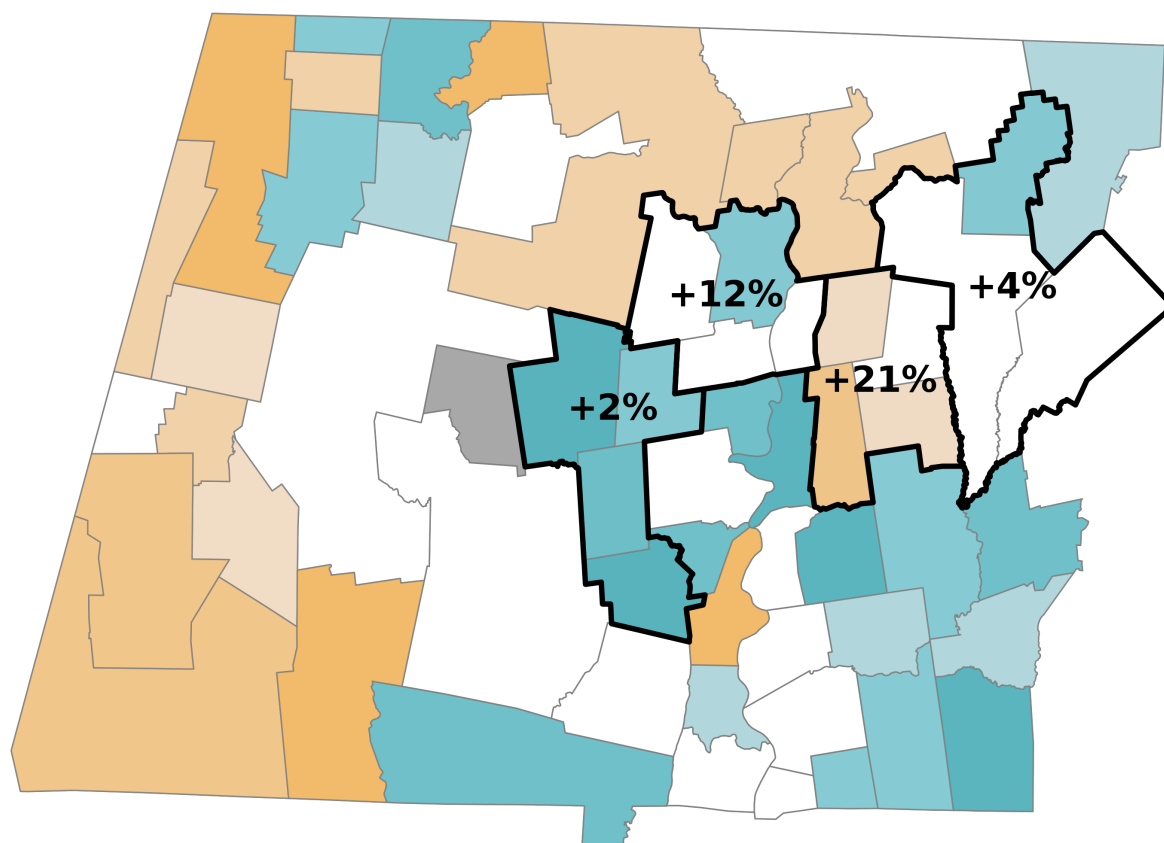


Figure 10

Geography matters for understanding school costs. This map shows where enrollment cohorts cluster and how regional district boundaries cross municipal lines.

Appendix D. Additional Visualizations (continued)

Geographic map showing 2009 PPE vs enrollment cohort baseline



Total PPE vs Cohort Baseline - 2009 ($\pm 5\%$ threshold)

■ $\leq -20\%$ cohort avg: 5 district(s)	■ +10 to +15% cohort avg: 8 district(s)
■ -15 to -20% cohort avg: 6 district(s)	■ +15 to +20% cohort avg: 3 district(s)
■ -10 to -15% cohort avg: 8 district(s)	■ $\geq +20\%$ cohort avg: 5 district(s)
■ -5 to -10% cohort avg: 5 district(s)	■ Missing data: 1 district(s)
□ Within $\pm 5\%$ cohort avg: 20 district(s)	+/-% = Secondary regional district deviation (n=4)
■ +5 to +10% cohort avg: 4 district(s)	

Figure 11

This map shows each district's 2024 per-pupil spending compared to its enrollment cohort baseline (weighted average of all Western MA districts in that cohort). This helps identify which districts are operating above or below the typical spending level for their enrollment size, accounting for the well-documented relationship between district size and per-pupil costs.